

Attention Scarcity and Optimal Depth in AI Delegation Chains

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Abstract

When AI agents delegate to other AI agents, the scarcity of attention—not the cost of coordination—bounds organizational depth. We embed rational inattention into a multi-layer hierarchy: each agent allocates scarce attention across upstream signals subject to a finite budget (context window). The endogenous transmission factor emerges from Bayesian updating rather than being assumed. We prove that optimal depth is finite and interior, that precision loss compounds geometrically through the chain, and that front-loaded budget allocation—larger context windows at early layers—maximizes output quality. When API costs vanish, the bound on depth persists because finite context windows ensure that each layer must compress information, and that compression compounds. We calibrate the model to LLM benchmark data and propose five testable predictions with specific regression specifications and identification strategies.

JEL Codes: D23, D82, L22, L86, O32, O33

1 Introduction

Over 60% of enterprise AI systems now employ multi-agent architectures, yet the optimal depth of these delegation chains remains theoretically unexplored. As context windows expand (from GPT-3’s 2K tokens to Gemini-1.5’s 1M tokens), the intuition is that deeper chains—more layers of specialized agents—should become optimal. We show that this intuition is wrong.

The reason is attention scarcity. Each agent in a chain receives signals from the layer below, but its context window is finite. It must choose which signals to attend to and which to ignore. The ignored signals are lost, and the attended signals are processed with reduced precision. This precision loss compounds through the chain, creating an endogenous bound on depth that persists even as API costs vanish.

Our model embeds rational inattention (Sims 2003) into a multi-layer information-processing hierarchy. The key innovation is the endogenous transmission factor ρ_ℓ , defined as the ratio of achieved posterior precision to first-best posterior precision. Unlike the exogenous distortion factor in previous work, ρ_ℓ emerges from the Bayesian updating problem and reflects the actual information degradation at each layer.

Contribution relative to classical frameworks. Our analysis is related to, but distinct from, the canonical models of multi-layer organization developed by Garicano (2000) and Holmström and Milgrom (1991) (henceforth, GHM). In the GHM framework, agents choose scalar effort levels subject to convex costs, and finite organizational depth arises from the interplay of diminishing returns and increasing marginal costs. We show in Section 4.9 that if one recasts the multi-agent reasoning problem in this classical form, several empirically relevant predictions are lost: optimal depth diverges as marginal costs vanish, budget allocation is uniform across layers, and signal heterogeneity has no effect on organizational performance. These predictions are restored only when one recognizes that the relevant economic margin is not scalar effort but *vector-valued attention allocation* under a binding budget constraint. Our model therefore bridges the classical theory of hierarchical organization with the information-theoretic approach to bounded rationality (Sims 2003), yielding a framework that is both analytically tractable and empirically disciplined by observable attention weights from modern AI systems.

This paper makes three contributions. First, we replace the exogenous transmission factor in Garicano (2000) and Dessein and Santos (2006) with a structurally derived object. The transmission factor ρ_ℓ is the ratio of achieved posterior precision to first-best precision, and it emerges from the optimal attention-allocation problem rather than being assumed. This connects the organizational model to the Bayesian micro-foundations of rational inattention (Sims 2003).

Second, we show that the limit to delegation is a technological constraint (the context window), not a financial one. Even as API costs fall to zero, the finite attention budget ensures that information is lost at each layer, and that loss compounds. This implies that Coasian arguments about the cost of coordination do not fully explain the boundaries of AI organizations: the bottleneck is attention, not transaction costs.

Third, we provide a calibration framework that maps the model’s parameters to directly observable LLM technology: context window lengths, API pricing, and benchmark accuracy. The calibration demonstrates that the key predictions—finite optimal depth, compounding precision loss, and the irrelevance of API costs for depth—hold for empirically plausible parameter values. We also provide empirical validation through regression analysis of benchmark data, confirming the model’s testable predictions.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the model. Section 4 analyzes the single-layer problem. Section 5 extends the analysis to multiple layers and derives the optimal depth. Section 6 calibrates the model. Section 7 discusses empirical implications and presents regression evidence. Section 8 considers extensions. Section 9 concludes.

2 Related Literature

Our work bridges four literatures: organizational economics, rational inattention, AI and organizations, and information design.

2.1 Organizations and Knowledge Hierarchies

The organizational economics literature on knowledge hierarchies and multi-layer contracting studies how information loss and control loss bound depth. Garicano (2000) provides the canonical model of knowledge-based hierarchies, where problems are routed to agents with appropriate expertise. Melumad, Mookherjee, and Reichelstein (1995)

analyze how control loss in hierarchies creates a trade-off between specialization and coordination. Lash and Wellington (2008) explicitly models control loss as a constraint on depth. Bolton and Dewatripont (1994) study the firm as a communication network. Radner (1993) analyze decentralized information processing. Our contribution is to microfound the distortion mechanism in attention allocation rather than assuming it.

2.2 Rational Inattention

The rational inattention literature studies how information-processing constraints shape decision-making. Sims (2003) introduces the rational inattention framework, where agents face an entropy constraint on information processing. Sims (2006) applies this to monetary policy. Mackowiak, Matějka, and Wiederholt (2023) provides a comprehensive review. Dessein and Santos (2006) study organizational attention in a two-layer setting, showing how attention scarcity shapes delegation. Dessein, Galeotti, and Santos (2016) extend this to communication in organizations. Caplin and Dean (2015) develop revealed-preference tests for rational inattention. Our model extends the organizational attention framework to arbitrarily deep chains.

2.3 Multi-Stage Information Design

A growing literature studies information design in dynamic and multi-stage settings. In these environments, a sender strategically controls the flow of information to a receiver (or receivers) who take actions over time. Our paper shares with this literature the fundamental premise that information is transmitted and evolves across multiple stages, but differs in emphasizing constrained information *processing*—arising from attention scarcity within hierarchical organizations—rather than strategic information *withholding* by a sender.

Ely (2017) studies a dynamic information transmission problem in which a sender communicates the state of a Markov process to a receiver via binary signals (“beeps”). The optimal signaling strategy involves selective disclosure: the sender pools states and releases information only when it is most consequential for the receiver’s decision. The key insight is that the *timing* of information revelation is a strategic choice, and the sender may optimally delay or suppress information to influence the receiver’s beliefs. Our framework differs in that the constraint on information transmission arises not from a sender’s strategic objectives but from the organizational limitation that each processing layer has finite attention capacity.

Renault, Solan, and Vieille (2017) analyze optimal dynamic information provision in a sender-receiver framework with a persistent binary state. They characterize the sender’s value function and show that the optimal policy involves gradual revelation: information is released piecemeal over time to maximize the sender’s expected payoff. The dynamics of belief updating are central to their analysis, as the sender faces a trade-off between revealing information early and withholding it to preserve future influence. In our model, similar intertemporal trade-offs arise, but they are driven by organizational rather than strategic considerations.

Escudé and Sinander (2023) study “slow persuasion,” in which a sender provides public information over time subject to a graduality constraint. They show that when information can only accumulate gradually, the sender’s ability to persuade is inhibited relative to the unconstrained benchmark. The graduality constraint in their model is

exogenous, much as the attention constraint in our framework is determined by the organizational technology. The key difference is that their constraint limits the *speed* of information acquisition, whereas ours limits the *volume* of information that can be processed at each layer.

Smolin (2021) studies a principal who designs a dynamic evaluation policy for an agent of uncertain productivity. Smolin shows that the equilibrium is Pareto efficient and that the wage deterministically grows with tenure. The evaluation policy serves as a dynamic information revelation mechanism. Our model shares the feature that information structures are chosen dynamically, but in our setting the designer chooses the *architecture* of information processing rather than the informativeness of signals per se.

The common thread across these papers is that information is a flow variable that must be managed over time. Our contribution is to study how this flow is governed by organizational structure: in a hierarchy with attention-constrained agents, the optimal architecture determines not only *what* information is transmitted but also *how much* of it survives at each stage.

2.4 AI and Organizations

A substantial empirical and theoretical literature documents the effects of artificial intelligence on productivity, labor markets, and the organization of work. Our paper complements this literature by examining the *internal* structure of AI systems—how multiple AI agents should be organized into hierarchical architectures—rather than the effects of AI on human labor markets.

Acemoglu and Restrepo (2018) develop a task-based framework in which automation technologies displace labor in tasks previously performed by workers. Their key insight is that while automation directly reduces labor demand, it also indirectly encourages the creation of new tasks in which labor has a comparative advantage. Acemoglu and Restrepo (2022) extend this framework to study wage inequality, showing that a substantial fraction of the rise in US wage inequality since 1980 can be attributed to automation of routine tasks. Our paper takes a different perspective: rather than studying how AI substitutes for or complements human labor in existing organizational structures, we ask how AI agents themselves should be organized into efficient architectures for processing information.

Acemoglu, Autor, Hazell, and Restrepo (2022) study the impact of AI on labor markets using evidence from online vacancies. They document that AI-related skills are increasingly demanded in occupations that combine cognitive and technical tasks, and that AI adoption is associated with shifts in skill requirements rather than pure displacement.

Noy and Zhang (2023) provide experimental evidence that generative AI tools substantially increase worker productivity. In a randomized controlled trial with professional writers, they find that access to ChatGPT reduces task completion time by 40% and increases output quality. Importantly, the productivity gains are largest for less-skilled workers, suggesting that AI can compress skill-based wage differentials. Peng et al. (2023) report similar findings for software developers using GitHub Copilot, documenting a 55.8% reduction in task completion time.

Ide and Talamas (2025) embed AI into the knowledge-hierarchy framework of Garicano (2000) but without attention scarcity. Klein and Wieczorek (2026) study AI-driven

organizational flattening but abstract from incentive constraints. We directly address the gap between AI capabilities (expanding context windows) and organizational limits (compounding attention loss).

Our work is complementary to this literature in the following sense. The existing literature studies the *macro* effects of AI adoption on human workers and firms—how AI changes the demand for skills, the distribution of wages, and the productivity of existing organizational forms. In contrast, we study the *micro* design of AI systems themselves: given that AI agents have heterogeneous capabilities and attention constraints, what is the optimal hierarchical architecture for information processing?

2.5 Information Design

The information design literature provides tools for modeling strategic communication. Kamenica and Gentzkow (2011) introduce Bayesian persuasion, where a sender designs a signal to influence a receiver’s beliefs. Lien, Qiu, and Zhong (2019) extend this to multi-layer communication networks. Our model differs in that the “design” is not strategic but constrained by attention scarcity.

Table 1: Literature Mapping

Contribution	Relation to Our Model	Differentiation
Garicano (2000) Dessein and Santos (2006)	Knowledge hierarchy Organizational attention	+ attention scarcity + continuous allocation + chain degradation
Hadfield-Menell (2021) Phelps and Ranson (2023) Yang and Zhai (2025)	Single-layer PA in AI Behavioral validation with LLMs Heuristic: coordination costs	+ chain extension + cascade mechanism + multi-layer experimental design Formal proof: alignment costs persist
Tomašev, Paquet, and Tuyls (2026)	Bounded rationality in chains	Incentive misalignment in chains
Klein and Wiczorek (2026)	Coordination flattening	Attention scarcity bounds depth

Table 1 summarizes how our work relates to the most closely connected papers in each literature.

3 Model Setup

3.1 Overview: From Effort Choice to Attention Allocation

The central premise of this paper is that AI agents in a delegation chain do not face a moral-hazard problem over effort; they face a rational-inattention problem over at-

tention. Each agent possesses a finite attention budget—directly interpretable as the context-window length or the number of attention heads in a transformer architecture—and must decide how to allocate that budget across a large number of upstream signals. Distortion arises not because agents shirk, but because scarcity forces them to ignore some signals. The optimal depth of the chain is determined by the point at which the marginal attention cost of adding another layer exceeds the marginal information gain from deeper processing.

This shift from effort to attention is not merely a relabeling. It changes the source of distortion, the nature of the optimization problem, and the comparative statics. In the classical principal-agent framework à la Holmström and Milgrom (1991), the agent chooses a scalar effort level e that maps into output through a production function $q = f(e) + \varepsilon$. The distortion is moral hazard: the agent would prefer to set $e = 0$. In our framework, the agent chooses a vector of attention weights $\alpha = (\alpha_1, \dots, \alpha_K)$ subject to a budget constraint $\sum_k \alpha_k \leq A$. The distortion is rational inattention: even if the agent fully exhausts the budget A , some signals receive zero weight and are therefore processed with error. The agent is not shirking; she is overwhelmed.

This distinction has three immediate implications for the study of AI delegation chains. First, it provides a direct micro-foundation: the attention budget A_ℓ at layer ℓ is the context-window length of the LLM (e.g., 4,096 tokens for GPT-3.5, 128,000 for GPT-4, or 1,000,000 for Gemini 1.5 Pro). The attention weights $\alpha_{\ell,k}$ map onto the attention-head activations in the transformer. The cost of attention $c_\ell(A_\ell)$ is the API pricing for longer contexts.

Second, it endogenizes the precision parameter. In the HM framework, the signal precision τ is an exogenous parameter that the principal might manipulate through incentives. In our framework, the effective precision τ_ℓ^* is derived from the optimal solution to the attention-allocation problem and depends on the number of signals, their intrinsic importance, and the attention budget—all of which are either endogenous choices or directly observable.

Third, it structures the output function. Equation (3) below derives precision from attention allocation. The production function is not assumed; it follows from the technology of selective attention.

Our framework builds on the rational-inattention literature initiated by Sims (2003, 2006) and its organizational extension by Dessein and Santos (2006) and Dessein, Galeotti, and Santos (2016). The key departure from Sims is that our agents operate in a hierarchy: the output of one agent’s attention-allocation problem becomes the input to the next agent’s problem. The key departure from Dessein–Santos is that our agents are artificial—their attention budgets, costs, and signal structures are directly measurable from API logs—which allows us to calibrate and test the model with unprecedented precision.

3.2 The Environment

Assumption 1 (Hierarchy). *There are $L + 1$ layers indexed by $\ell = 0, 1, \dots, L$. Layer 0 is the principal (human or top-level AI) who sets the organizational architecture. Layers $\ell = 1, \dots, L$ are AI agents arranged in a chain: the output of layer $\ell - 1$ serves as the input to layer ℓ . Layer L is the final executor that produces the observable output.*

Assumption 2 (State and Signals). *The underlying state is $\omega \in \Omega \subseteq \mathbb{R}^d$, representing the correct response to a complex task (e.g., the correct answer to a coding problem, the*

accurate diagnosis from medical records, or the valid proof of a mathematical theorem). At each layer ℓ , the agent receives K_ℓ signals $\{s_{\ell,1}, \dots, s_{\ell,K_\ell}\}$ from the layer above. Each signal is a noisy projection of the state:

$$s_{\ell,k} = H_{\ell,k}\omega + \varepsilon_{\ell,k}, \quad \varepsilon_{\ell,k} \sim \mathcal{N}(0, \sigma_{\ell,k}^2 I), \quad (1)$$

where $H_{\ell,k} \in \mathbb{R}^{d_k \times d}$ is a linear observation operator and $\sigma_{\ell,k}^2$ is the intrinsic noise variance. The intrinsic precision of signal k at layer ℓ is $\tau_{\ell,k}^0 \equiv 1/\sigma_{\ell,k}^2$.

The signal structure in Assumption 2 captures the empirical reality of AI delegation chains. In a retrieval-augmented generation (RAG) pipeline, the signals are retrieved documents; in a multi-agent coding system, they are code snippets and error messages from upstream agents; in a hierarchical reasoning chain, they are intermediate conclusions. The observation operator $H_{\ell,k}$ captures the fact that each signal may reveal only a subset of the state.

Remark 1 (Scalar-State Simplification). *In the baseline analysis, we set $d = 1$ (scalar state) and $H_{\ell,k} = 1$ for all ℓ, k . This simplifies the precision-aggregation technology to scalar addition: $\tau_\ell = \tau_\ell^{\text{prior}} + \sum_k g_\ell(\alpha_{\ell,k})\tau_{\ell,k}^0$. The scalar assumption is restrictive when signals reveal different dimensions of a multi-faceted state. In Section 8, we sketch how the model extends to $d > 1$ with a precision matrix; the key insight—that attention scarcity bounds depth through compounding precision loss—is preserved.*

Assumption 3 (Attention Budget). *Each agent at layer ℓ has a finite attention budget $A_\ell > 0$. The agent chooses attention weights $\alpha_\ell = (\alpha_{\ell,1}, \dots, \alpha_{\ell,K_\ell})$ satisfying*

$$\sum_{k=1}^{K_\ell} \alpha_{\ell,k} \leq A_\ell, \quad \alpha_{\ell,k} \geq 0 \quad \forall k. \quad (2)$$

The weight $\alpha_{\ell,k}$ measures the fraction of the agent’s attention capacity devoted to signal k . If $\alpha_{\ell,k} = 0$, the signal is completely ignored.

The attention budget A_ℓ is the central scarce resource in our model. In transformer-based LLMs, it has three direct interpretations: (i) the context-window length; (ii) the number of attention heads; and (iii) the effective working-memory capacity. The budget constraint (2) is hard: exceeding A_ℓ causes the model to truncate, compress, or drop signals.

It is important to emphasize that A_ℓ is not a choice variable of the agent at layer ℓ ; it is a technological parameter determined by the model architecture. The principal chooses A_ℓ when selecting which model to deploy at each layer. This creates a two-stage organizational design problem: the principal chooses the architecture $\{A_\ell\}_{\ell=1}^L$ and the chain depth L ; the agents then solve their attention-allocation problems given these budgets.

Assumption 4 (Attention Technology). *The effective precision achieved by the agent at layer ℓ is*

$$\tau_\ell(\alpha_\ell) = \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k})\tau_{\ell,k}^0, \quad (3)$$

where $g_\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is a twice-differentiable, increasing, and concave function satisfying $g_\ell(0) = 0$, $g'_\ell(0) > 0$, and $\lim_{\alpha \rightarrow \infty} g'_\ell(\alpha) = 0$.

The function g_ℓ captures the attention technology: how does allocating attention weight α to a signal translate into effective processing precision? The concavity of g_ℓ reflects diminishing returns to attention. Our baseline specification uses a power function:

$$g_\ell(\alpha) = \alpha^\gamma, \quad \gamma \in (0, 1). \quad (4)$$

The parameter γ is the attention elasticity of precision. When γ is close to 1, attention is nearly linear in precision. When γ is small, attention is highly concave. We calibrate γ from empirical data on LLM accuracy as a function of context length in Section 6.

Assumption 5 (Cost of Attention). *The cost of maintaining attention capacity A_ℓ at layer ℓ is $c_\ell(A_\ell)$, where $c_\ell : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is twice-differentiable, increasing, and convex, with $c_\ell(0) = 0$ and $c'_\ell(0) > 0$.*

For our baseline analysis, we use a quadratic cost:

$$c_\ell(A) = \kappa_\ell A + \frac{\phi_\ell}{2} A^2, \quad (5)$$

where $\kappa_\ell > 0$ is the marginal cost of the first unit of attention and $\phi_\ell \geq 0$ captures the convexity of compute costs at scale.

3.3 The Agent’s Problem: Attention Allocation

Given the environment above, the agent at layer ℓ solves the following optimization problem:

$$\max_{\boldsymbol{\alpha}_\ell : (2)} V_\ell(\tau_\ell(\boldsymbol{\alpha}_\ell)) - c_\ell(A_\ell), \quad (6)$$

where $V_\ell(\tau)$ is the value of posterior precision τ for decision-making at layer ℓ .

The value function $V_\ell(\tau)$ depends on the decision problem facing the agent. In our baseline, we assume quadratic loss (normal–normal conjugate model), which yields:

$$V_\ell(\tau) = -\psi_\ell \text{Var}(\omega \mid \tau) = -\frac{\psi_\ell}{\tau}, \quad (7)$$

where $\psi_\ell > 0$ is the marginal value of accuracy. With normal signals and a normal prior, the posterior variance is $\text{Var}(\omega \mid \tau) = 1/\tau$, so $V_\ell(\tau) = -\psi_\ell/\tau$. Since τ enters negatively, maximizing V_ℓ is equivalent to maximizing τ .

More generally, $V_\ell(\tau)$ can incorporate non-quadratic payoffs, discrete actions, or strategic interactions with downstream agents. We explore these extensions in Section 8. For the single-layer analysis in Section 4, the precise functional form of V_ℓ matters only through its monotonicity and curvature.

3.4 Key Distinctions from the Garicano–Holmström–Milgrom Framework

Before proceeding to the analysis, we pause to articulate three core distinctions between our attention-allocation framework and the GHM framework.

Distinction 1: Attention, not effort, is the scarce input. In the HM model, agents choose costly effort e to improve output quality. The distortion is moral hazard: agents prefer low effort, and incentives must be designed to induce the efficient level. In our model, agents choose attention weights α to improve signal precision. The distortion is rational inattention: agents cannot attend to everything, and the organizational design must determine which signals receive attention and which are deliberately ignored. This maps directly onto the empirical reality of LLM pipelines: the bottleneck is not the model’s “effort” but its “attention” (longer prompts require more context processing, and some information is necessarily dropped).

Distinction 2: Selective ignorance, not shirking, generates distortion. In the HM model, distortion arises because the agent’s effort is unobservable and the agent under-supplies it relative to the first-best. In our model, distortion arises because the agent rationally ignores some signals. The ignored signals are not random; they are the ones with low intrinsic precision, low importance for the decision, or high attention cost. This generates testable predictions: in an LLM pipeline, we should observe that models drop low-quality retrieved documents before high-quality ones, and that the dropout rate increases with the number of documents.

Distinction 3: Organization design as attention architecture, not incentive design. In the HM model, the principal designs incentives (wages, bonuses, monitoring) to align the agent’s effort with organizational goals. In our model, the principal designs the attention architecture: which signals flow to which layers, how much context each layer receives, and when to aggregate vs. specialize. The design variables are $\{A_\ell\}$ (attention budgets), $\{K_\ell\}$ (signal counts), and L (chain depth)—not wages or performance contracts.

3.5 Summary of Notation

For reference, Table 2 summarizes the key symbols used in the model.

4 Single-Layer Analysis

In this section, we solve the attention-allocation problem for a single layer, derive the optimal attention weights, characterize the resulting distortion, and establish comparative statics with respect to the attention budget, signal importance, and signal heterogeneity. We then connect our results to the rational-inattention literature and derive two propositions that form the building blocks for the multi-layer analysis in Section 5.

The single-layer problem is not merely a stepping stone. It is the core economic mechanism of the paper: everything that happens in a delegation chain—distortion, error propagation, depth optimization—traces back to how each individual layer allocates its scarce attention.

4.1 The Optimization Problem

Consider layer ℓ in isolation. The agent faces K_ℓ signals with intrinsic precisions $\{\tau_{\ell,k}^0\}_{k=1}^{K_\ell}$ and chooses attention weights $\alpha_\ell = (\alpha_{\ell,1}, \dots, \alpha_{\ell,K_\ell})$ subject to the budget constraint (2).

Table 2: Notation Summary

Symbol	Definition
ℓ	Layer index, $\ell = 0, 1, \dots, L$
ω	Underlying state (correct answer / true state)
$s_{\ell,k}$	Signal k at layer ℓ
K_ℓ	Number of signals at layer ℓ
$\tau_{\ell,k}^0$	Intrinsic precision of signal (ℓ, k)
A_ℓ	Attention budget at layer ℓ
$\alpha_{\ell,k}$	Attention weight on signal (ℓ, k)
$g_\ell(\cdot)$	Attention technology (precision transformation)
$\tau_\ell(\boldsymbol{\alpha}_\ell)$	Effective precision at layer ℓ
$V_\ell(\tau)$	Value of precision τ for decision-making
$c_\ell(A_\ell)$	Cost of attention capacity A_ℓ
γ	Attention elasticity of precision
κ_ℓ, ϕ_ℓ	Cost parameters (linear and quadratic)
$\delta_{\ell,k}$	Distortion on signal k at layer ℓ
ρ_ℓ	(Simplified) Distortion transmission factor

The effective precision is given by (3), and the agent's payoff is $V_\ell(\tau_\ell(\boldsymbol{\alpha}_\ell)) - c_\ell(A_\ell)$. Since $c_\ell(A_\ell)$ does not depend on $\boldsymbol{\alpha}_\ell$, the optimization problem is:

$$\max_{\boldsymbol{\alpha}_\ell : (2)} V_\ell \left(\sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0 \right). \quad (8)$$

We assume throughout that $V_\ell(\tau)$ is increasing and concave in τ . This is satisfied by the quadratic-loss specification (7) and by most standard decision problems.

4.2 First-Order Conditions and Optimal Attention

Let $\lambda_\ell \geq 0$ be the Lagrange multiplier on the budget constraint (2). The Lagrangian is:

$$\mathcal{L}(\boldsymbol{\alpha}_\ell, \lambda_\ell) = V_\ell \left(\sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0 \right) - \lambda_\ell \left(\sum_{k=1}^{K_\ell} \alpha_{\ell,k} - A_\ell \right) + \sum_{k=1}^{K_\ell} \mu_{\ell,k} \alpha_{\ell,k}, \quad (9)$$

where $\mu_{\ell,k} \geq 0$ are multipliers on the non-negativity constraints.

The first-order condition for $\alpha_{\ell,k}$ is:

$$V'_\ell(\tau_\ell(\boldsymbol{\alpha}_\ell)) \cdot g'_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0 = \lambda_\ell - \mu_{\ell,k}. \quad (10)$$

This condition has a straightforward economic interpretation. The left-hand side is the marginal benefit of allocating attention to signal k : the marginal value of precision (V'_ℓ) times the marginal precision gain from attention ($g'_\ell(\alpha_{\ell,k}) \tau_{\ell,k}^0$). The right-hand side is the marginal cost: the shadow price of the attention budget (λ_ℓ) minus the relaxation of the non-negativity constraint ($\mu_{\ell,k}$). At an interior solution ($\alpha_{\ell,k} > 0$), $\mu_{\ell,k} = 0$ and the marginal benefit equals the shadow price λ_ℓ .

Proposition 1 (Optimal Attention Allocation). *Suppose V_ℓ is increasing and concave, g_ℓ is increasing and concave with $g'_\ell(0) > 0$, and $\tau_{\ell,k}^0 > 0$ for all k . Then:*

(i) The optimal attention allocation $\alpha_\ell^*(A_\ell)$ exists and is unique.

(ii) If the budget binds ($\sum_k \alpha_{\ell,k}^* = A_\ell$), the optimal weights satisfy

$$g'_\ell(\alpha_{\ell,k}^*)\tau_{\ell,k}^0 = g'_\ell(\alpha_{\ell,j}^*)\tau_{\ell,j}^0 \quad \forall k, j \text{ with } \alpha_{\ell,k}^*, \alpha_{\ell,j}^* > 0. \quad (11)$$

(iii) The optimal weight on signal k is increasing in its intrinsic precision $\tau_{\ell,k}^0$ and decreasing in the precision of other signals.

Proof. See Appendix A. $\square$

Intuition. The optimality condition (11) says that the agent equalizes the marginal precision product across all signals that receive positive attention. Signals with higher intrinsic precision receive more attention not because they are “preferred” in any absolute sense, but because they yield higher marginal returns: each unit of attention devoted to a high-quality signal extracts more information than a unit devoted to a low-quality signal. This is the familiar logic of equalizing marginal products across inputs, applied here to the allocation of attention rather than labor or capital.

For the power-function specification $g_\ell(\alpha) = \alpha^\gamma$, the first-order condition becomes:

$$\alpha_{\ell,k}^* = \left(\frac{\gamma \tau_{\ell,k}^0}{\lambda_\ell} \right)^{\frac{1}{1-\gamma}} \quad (\text{interior solutions}). \quad (12)$$

This closed form reveals the key comparative static: the optimal attention weight is log-convex in intrinsic precision when $\gamma < 1/2$ and log-concave when $\gamma > 1/2$. In the empirically relevant case of $\gamma \approx 0.3$ (calibrated from LLM accuracy data), small differences in signal quality are amplified into large differences in attention allocation: a signal with twice the precision receives approximately $2^{1/(1-0.3)} \approx 2.69$ times the attention. This amplification is the mechanism by which “important” signals dominate and “unimportant” signals are starved.

4.3 Distortion and the Attention Gap

We now define the distortion arising from attention allocation and characterize how it depends on the attention budget and signal structure.

Definition 1 (Attentional Distortion). *The distortion on signal k at layer ℓ is*

$$\delta_{\ell,k} = (\bar{\alpha}_{\ell,k} - \alpha_{\ell,k}^*)^+, \quad (13)$$

where $\bar{\alpha}_{\ell,k}$ is the attention weight required to fully process signal k (i.e., achieve $\tau_{\ell,k}^0$ without loss), and $(\cdot)^+ = \max\{\cdot, 0\}$.

In the baseline where $\bar{\alpha}_{\ell,k} = 1$ for all k (unit attention achieves full precision), the distortion is simply $\delta_{\ell,k} = (1 - \alpha_{\ell,k}^*)^+$. If $\alpha_{\ell,k}^* \geq 1$, the signal is fully processed and $\delta_{\ell,k} = 0$. If $\alpha_{\ell,k}^* < 1$, the signal is under-processed and the distortion is positive.

We can also define an aggregate distortion measure:

Definition 2 (Aggregate Distortion). *The aggregate attentional distortion at layer ℓ is*

$$\Delta_\ell = \frac{\sum_{k=1}^{K_\ell} \omega_{\ell,k} \delta_{\ell,k}}{\sum_{k=1}^{K_\ell} \omega_{\ell,k}}, \quad (14)$$

where $\omega_{\ell,k} > 0$ is the importance weight of signal k for the layer’s decision. In the quadratic-loss case, $\omega_{\ell,k} = \tau_{\ell,k}^0$.

Proposition 2 (Distortion Comparative Statics). *The aggregate distortion Δ_ℓ satisfies:*

- (i) Δ_ℓ is decreasing in the attention budget A_ℓ : $\partial\Delta_\ell/\partial A_\ell < 0$.
- (ii) Δ_ℓ is increasing in the number of signals K_ℓ : $\partial\Delta_\ell/\partial K_\ell > 0$ when A_ℓ is held fixed.
- (iii) Δ_ℓ is decreasing in the heterogeneity of signal precisions: holding mean precision fixed, Δ_ℓ is lower when $\{\tau_{\ell,k}^0\}$ is more dispersed.

Proof. See Appendix B. □

Intuition. Proposition 2 delivers three testable predictions. First, increasing the attention budget reduces distortion: if the principal upgrades from GPT-3.5 (4K context) to GPT-4 (128K context), the model can attend to more signals and distortion falls. Second, increasing the number of signals while holding the budget fixed increases distortion: this is the mechanism behind “attention overload” in RAG systems. Third, signal heterogeneity reduces aggregate distortion because the optimal allocation concentrates attention on the best signals. If all signals are of equal quality, the agent spreads attention evenly and all signals are under-processed. If signals vary in quality, the agent concentrates on the high-quality ones, achieving near-full precision on the most important signals while deliberately sacrificing the low-quality ones.

4.4 The Endogenous Distortion Factor

We now derive the endogenous distortion measure that connects the single-layer attention-allocation problem to the multi-layer transmission analysis.

Definition 3 (Single-Layer Distortion Measure). *The single-layer distortion measure at layer ℓ is*

$$\tilde{\delta}_\ell = 1 - \Delta_\ell = 1 - \frac{\sum_k \omega_{\ell,k} (\bar{\alpha}_{\ell,k} - \alpha_{\ell,k}^*)^+}{\sum_k \omega_{\ell,k}}. \quad (15)$$

The measure $\tilde{\delta}_\ell \in [0, 1]$ captures the fraction of signal importance that survives the attention-allocation process at a single layer, ignoring upstream effects. When $\tilde{\delta}_\ell = 1$, all signals are fully processed and no distortion occurs. When $\tilde{\delta}_\ell = 0$, all signals are ignored and the layer transmits pure noise. In general, $\tilde{\delta}_\ell$ is a function of the attention budget, the number of signals, their precisions, and the attention technology:

$$\tilde{\delta}_\ell = \tilde{\delta}(A_\ell, K_\ell, \{\tau_{\ell,k}^0\}, g_\ell). \quad (16)$$

This is the core endogenization result. The distortion measure is not a primitive parameter but a derived object that depends on observable technological and organizational variables.

Remark 2 (From $\tilde{\delta}_\ell$ to ρ_ℓ). *The single-layer measure $\tilde{\delta}_\ell$ is useful for analyzing distortion at an isolated layer. In the multi-layer analysis (Section 5), we introduce the endogenous transmission factor $\rho_\ell \equiv \tau_\ell^*/\tau_\ell^{FB}$, which accounts for the compounding of inherited precision. For the bottom layer with $\tau_0^{\text{prior}} = 0$, the two coincide: $\rho_0 = \tilde{\delta}_0 = 1 - \Delta_0$. For upper layers, ρ_ℓ generalizes $\tilde{\delta}_\ell$ by incorporating the prior precision.*

4.5 Connection to Rational Inattention

Our model is a direct extension of the rational-inattention framework to hierarchical organizations. We now make this connection explicit.

Sims (2003): The single-agent benchmark. In Sims’s model, a single agent faces a multidimensional state $\omega \in \mathbb{R}^d$ and chooses an information structure subject to a mutual-information constraint $I(\omega; \hat{\omega}) \leq \kappa$. Our model specializes Sims in two directions and generalizes it in one. We specialize by assuming a linear precision-aggregation technology (Equation (3)) rather than the general information-structure choice. We specialize by assuming a hard attention budget rather than an entropy-based information-cost function. We generalize by embedding the single-agent problem in a hierarchy where the output of one agent becomes the input to the next.

The hard budget vs. entropy-cost distinction is important. Sims’s entropy cost is analytically elegant, but it is less directly mapped to AI technology. The mutual information $I(\omega; \hat{\omega})$ is difficult to measure in an LLM pipeline, whereas the context-window length A_ℓ is directly observable. Our hard-budget formulation trades some analytical elegance for empirical tractability.

Dessein and Santos (2006): Organizational attention. Dessein and Santos model an organization in which agents have a fixed attention capacity that can be used for both sending and receiving information. Our model extends theirs in three ways. First, we allow attention to be continuously allocated across signals rather than discretely assigned to tasks. Second, we allow the attention budget to vary across layers (A_ℓ need not equal $A_{\ell'}$), capturing the fact that different LLMs in a chain have different context-window sizes. Third, we model the endogenous degradation of information through the chain.

Dessein, Galeotti, and Santos (2016): Division of labor. In their follow-up paper, Dessein et al. study how attention constraints generate specialization. Our Proposition 2(iii) echoes this result: heterogeneity in signal quality leads to specialization in attention allocation. We extend their analysis by deriving the optimal chain depth from attention costs.

4.6 Alternative: Entropy Cost Formulation

4.6.1 Entropy Cost Setup

In the rational inattention framework of Sims (2003), the agent’s information processing cost is proportional to the reduction in uncertainty (measured by mutual information) achieved by observing a signal. Specifically, if the agent chooses an information structure $P(\tilde{s} | \omega)$ that maps the true state $\omega \sim \mathcal{N}(\mu, \tau^{-1})$ into a signal realization $\tilde{s}$, the associated cost is:

$$C(I) = \theta \cdot I(\omega; \tilde{s}), \quad (17)$$

where $\theta > 0$ is the marginal price of information and $I(\omega; \tilde{s})$ denotes the mutual information:

$$I(\omega; \tilde{s}) = h(\omega) - h(\omega | \tilde{s}) = \frac{1}{2} \log_2 \left(\frac{\tau_{\text{post}}}{\tau} \right), \quad (18)$$

where $\tau_{\text{post}} = \tau + \tau_s$ is the posterior precision after observing signal $\tilde{s} = \omega + \varepsilon$ with noise precision τ_s , and $h(\cdot)$ denotes differential entropy.

When a chain of L agents each face entropy costs, agent ℓ chooses a signal structure $P(\tilde{s}_\ell | \omega)$ at cost $\theta_\ell \cdot I(\omega; \tilde{s}_\ell)$, and then passes a processed message m_ℓ to the next layer.

The key difference from the hard-budget formulation is that *all signals are processed, but with endogenously varying precision*, rather than a subset being selected with perfect precision.

4.6.2 Entropy-Cost Version of Our Model

Consider our multi-layer information processing chain under entropy costs. Each layer ℓ chooses the precision τ_ℓ of its signal subject to an entropy cost $\theta_\ell \cdot I(\omega; \tilde{s}_\ell)$. The total cost for the chain is:

$$C_{\text{total}} = \sum_{\ell=1}^L \theta_\ell \cdot I(\omega; \tilde{s}_\ell) = \sum_{\ell=1}^L \frac{\theta_\ell}{2} \log \left(1 + \frac{\tau_\ell}{\tau_{\ell-1}^*} \right), \quad (19)$$

where $\tau_{\ell-1}^*$ is the effective precision inherited from the previous layer. The optimization problem at each layer is to choose τ_ℓ to maximize:

$$U_\ell(\tau_\ell) = V(\tau_\ell^*) - \theta_\ell \cdot I(\omega; \tilde{s}_\ell), \quad (20)$$

where $\tau_\ell^* = \tau_{\ell-1}^* + \tau_\ell$ is the posterior precision.

Key Result: Robustness of Finite Depth. We now establish that the finite-optimal-depth result carries over to the entropy-cost setting:

Proposition 3 (Finite Optimal Depth under Entropy Costs). *Suppose each layer ℓ faces an entropy cost with parameter $\theta_\ell > 0$, and the value function $V(\tau)$ satisfies $V'(\tau) > 0$, $V''(\tau) < 0$, and the Inada-like condition $\lim_{\tau \rightarrow \infty} V'(\tau) = 0$. Then:*

- (i) **Concave precision response:** *The optimal precision $\tau_\ell^*(\theta_\ell)$ chosen at each layer satisfies $\frac{\partial^2 \tau_\ell^*}{\partial \theta_\ell^2} > 0$, i.e., precision decreases convexly in the cost parameter.*
- (ii) **Diminishing information increment:** *The marginal contribution of layer ℓ to total precision, $\Delta_\ell \equiv \tau_\ell^* - \tau_{\ell-1}^*$, is decreasing in ℓ when θ_ℓ is non-decreasing.*
- (iii) **Finite optimal depth:** *If $\inf_\ell \theta_\ell > 0$, there exists a finite $L^* < \infty$ such that adding layer $L^* + 1$ yields negative net value.*

Proof Sketch. (i) At layer ℓ , the first-order condition for optimal precision is:

$$V'(\tau_{\ell-1}^* + \tau_\ell) = \frac{\theta_\ell}{2} \cdot \frac{1}{\tau_{\ell-1}^* + \tau_\ell}. \quad (21)$$

This implicitly defines τ_ℓ^* as a function of θ_ℓ . By the implicit function theorem and the convexity of the entropy cost in precision, the response function is concave in $1/\theta_\ell$.

(ii) Since τ_ℓ^* is increasing in inherited precision $\tau_{\ell-1}^*$ but with slope less than 1 (due to $V'' < 0$), the increment Δ_ℓ decreases along the chain when cost parameters are bounded away from zero.

(iii) The net value of adding layer $L + 1$ is $V(\tau_{L+1}^*) - V(\tau_L^*) - \theta_{L+1} \cdot I(\omega; \tilde{s}_{L+1})$. Since $V'(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$ while the marginal entropy cost remains bounded below by $\underline{\theta} \cdot (2\tau)^{-1} > 0$, there exists a finite threshold beyond which the cost exceeds the benefit. $\square$

Table 3: Comparison: Hard-Budget vs. Entropy-Cost Assumptions

Dimension	Hard-Budget	Entropy-Cost
Information loss	Signals beyond budget $\bar{A}_\ell$ ignored or truncated.	All signals processed with endogenously varying precision.
Optimization variable	Attention weight vector $\alpha_\ell \in \Delta^K$ (discrete capacity).	Information structure $P(\tilde{s} \mid \omega)$ (continuous precision).
Observability	$\bar{A}_\ell$ directly observable (context window size).	θ_ℓ latent; requires calibration.
LLM mapping	Direct: context window imposes hard token bound.	Indirect: entropy reduction to FLOPs mapping.
Core conclusion	Finite optimal depth (Prop. 12).	Finite optimal depth (Prop. 3).
Parametrization	Sharp cutoff at $\bar{A}_\ell$.	Smooth; no cutoff.
Applicable scenarios	Fixed context-window LLMs.	Variable-compute systems.

Intuition for Robustness. The critical insight is that the entropy cost function’s *convexity* in precision (arising from the logarithmic form of mutual information) naturally generates the same diminishing-returns property that our hard-budget model captures through the fixed-capacity constraint.

4.6.3 Discussion: Why Hard Budgets in the Baseline Model

We adopt the hard-budget assumption as our baseline for three substantive reasons related to *empirical operability* and the specific application to large language models:

Direct observability. The hard budget $\bar{A}_\ell$ maps directly and transparently to a measurable technological parameter: the context window length of an LLM (e.g., 4,096 tokens for GPT-3, 128,000 tokens for GPT-4 Turbo). This observable anchor allows us to discipline the model with empirical moments and conduct counterfactuals about window-size expansion.

Descriptive accuracy for current LLM architectures. Current production LLMs operate with fixed context windows: tokens beyond the window length are either truncated entirely or compressed through external summarization mechanisms. This is precisely the hard-budget mechanism.

Analytical tractability. The hard-budget formulation yields a finite-dimensional optimization over the simplex Δ^K , enabling closed-form solutions for the attention allocation and sharp comparative statics.

Entropy costs as robustness check. We view the entropy-cost formulation primarily as a *robustness check* on the qualitative conclusions. As Proposition 3 demonstrates, the core mechanism—diminishing returns to depth driven by concave precision accumulation—is preserved across both formulations.

4.7 Propositions for the Multi-Layer Analysis

We conclude the single-layer analysis with two propositions that will serve as building blocks for the multi-layer analysis.

Proposition 4 (Precision Response to Budget). *Let $\tau_\ell^*(A_\ell) \equiv \tau_\ell(\alpha_\ell^*(A_\ell))$ be the optimal effective precision. Then:*

- (i) $\tau_\ell^*(A_\ell)$ is increasing and concave in A_ℓ .
- (ii) The marginal return to attention, $\partial\tau_\ell^*/\partial A_\ell$, is decreasing in K_ℓ .
- (iii) $\tau_\ell^*(A_\ell)$ is bounded above by $\bar{\tau}_\ell \equiv \sum_k \tau_{\ell,k}^0$.

Proof. See Appendix C. □

Proposition 5 (Cost-Minimizing Budget). *For a target precision level $\bar{\tau}$, the cost-minimizing attention budget $A_\ell^*(\bar{\tau})$ satisfies:*

- (i) $A_\ell^*(\bar{\tau})$ is increasing and convex in $\bar{\tau}$.
- (ii) The marginal cost of precision, $\partial c_\ell(A_\ell^*)/\partial \bar{\tau}$, is increasing in K_ℓ .
- (iii) If $\bar{\tau} > \tau_\ell^*(K_\ell)$, then $A_\ell^*(\bar{\tau})$ does not exist: the target is infeasible.

Proof. See Appendix D. □

4.8 Discussion: Why This Is Not “Concave Benefit + Convex Cost”

We close this section by addressing a potential objection: is the attention-allocation model merely a relabeled version of the “concave benefit + convex cost = finite optimum” structure? We argue that it is not, for three reasons.

First, the concavity emerges from the technology, not the objective. In the standard framework, the principal assumes a concave benefit function and a convex cost function to ensure a finite optimum. In our framework, the concavity of $\tau_\ell^*(A_\ell)$ (Proposition 4) is *derived* from the attention technology: diminishing returns arise because the agent allocates attention to signals in order of quality, not because we assumed a concave production function.

Second, the optimization is over a vector, not a scalar. The HM model optimizes over a scalar effort e . Our model optimizes over a vector α_ℓ subject to a budget constraint. The solution is a system of K_ℓ equations with non-negativity constraints and a binding budget. The structure of the solution—which signals receive attention, which are ignored, and how the allocation changes with parameters—is economically meaningful and empirically testable.

Third, the distortion is endogenous and structurally derived. In the standard framework, the distortion parameter is exogenous and the output function is ad hoc. In our framework, the distortion measure $\tilde{\delta}_\ell$ is derived from the optimal attention allocation (Definition 3), the output function is the precision-aggregation technology (Equation (3)), and both are directly connected to observable AI technology.

4.9 Benchmark Model: The GHM Framework

To isolate the distinctive predictions of our attention-allocation model, we now construct and solve a benchmark formulation that nests our setting within the classical framework of Garicano (2000) and Holmström and Milgrom (1991) (henceforth, GHM).

4.9.1 Model Setup

Consider a chain of L identical agents indexed by $\ell = 1, \dots, L$. Each agent ℓ chooses a scalar effort level $e_\ell \in \mathbb{R}_+$, which represents the precision or quality of the signal that agent ℓ transmits upstream.

Assumption 6 (GHM Cost Structure). *The cost function ψ is twice continuously differentiable with:*

- (i) $\psi(0) = 0$, $\psi'(e) > 0$ for all $e \geq 0$;
- (ii) $\psi''(e) > 0$ for all $e \geq 0$ (strict convexity);
- (iii) $\lim_{e \rightarrow \infty} \psi'(e) = \infty$ (Inada condition).

Assumption 7 (GHM Production Technology). *The final output quality is given by*

$$q_L = f\left(\sum_{\ell=1}^L e_\ell\right) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (22)$$

where $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is twice continuously differentiable with $f'(x) > 0$, $f''(x) < 0$ for all $x \geq 0$, and satisfies the Inada conditions $f'(0) = \infty$, $\lim_{x \rightarrow \infty} f'(x) = 0$.

Agent ℓ 's utility is:

$$U_\ell = \mathbb{E}[w(q_L)] - \psi(e_\ell) = a + b \cdot f\left(\sum_{j=1}^L e_j\right) - \psi(e_\ell). \quad (23)$$

4.9.2 Equilibrium Effort and Optimal Depth

Taking the depth L and incentive power b as given, each agent ℓ chooses e_ℓ to maximize (23). By symmetry, $e_\ell = e^*$ for all ℓ . The first-order condition is:

$$b \cdot f'(L \cdot e^*) = \psi'(e^*). \quad (24)$$

Proposition 6 (GHM Optimal Depth). *Suppose Assumptions 6–7 hold and the principal optimally sets $b^* > 0$. Then:*

- (i) *There exists a finite optimal depth $L_{GHM}^* < \infty$.*
- (ii) *As the marginal cost of effort vanishes—formally, as $\psi'(e) \rightarrow 0$ —the optimal depth diverges:*

$$\lim_{\kappa \rightarrow 0} L_{GHM}^*(\kappa) = \infty. \quad (25)$$

- (iii) *The equilibrium effort per layer satisfies $e^* \rightarrow \infty$ as $\kappa \rightarrow 0$.*

Proof. (i) At $L = 0$, the marginal benefit is strictly positive because $f'(0) = \infty$. As $L \rightarrow \infty$, $f'(L \cdot e^*) \rightarrow 0$ while $\psi(e^*) > 0$. By continuity, there exists at least one crossing.

(ii) Consider the family $\psi_\kappa(e) = \kappa \cdot \tilde{\psi}(e)$. For any fixed finite L , as $\kappa \rightarrow 0$, the FOC implies $f'(L \cdot e^*) = (\kappa/b)\tilde{\psi}'(e^*) \rightarrow 0$, which requires $L \cdot e^* \rightarrow \infty$. The marginal benefit is $O(1)$ while the marginal cost is $O(\kappa)$. Thus the principal always wants to add more layers, and $L_{\text{GHM}}^* \rightarrow \infty$. $\square$

4.9.3 Contrast with Our Model

In our attention-allocation model, even as the per-unit cost of attention $\kappa \rightarrow 0$, each layer faces a *hard budget constraint* $\sum_k \alpha_{\ell,k} \leq A_\ell$. The optimal depth remains finite because:

- (a) The budget A_ℓ places an upper bound on effective precision.
- (b) The transmission factor $\rho_\ell = \tau_\ell^*/\tau_\ell^{\text{FB}} < 1$ is bounded away from 1 by the geometry of the budget constraint, regardless of κ .
- (c) The geometric decay of precision along the chain is driven by the *relative* allocation across signals, not by the absolute cost of attention.

Table 4: Model Predictions: Our Framework vs. GHM Benchmark

Prediction	Our Model	GHM Benchmark	Empirical Test
Finite optimal depth ($\kappa \rightarrow 0$)	Yes. Budget A_ℓ bounds precision regardless of cost.	No. $L_{\text{GHM}}^* \rightarrow \infty$ as $\psi'(e) \rightarrow 0$.	Observe depth response to API price cuts.
Budget allocation	Front-loaded. Attention concentrates at early layers.	Uniform. Identical effort e^* at all layers.	Compare model sizes across layers.
Signal heterogeneity	Reduces distortion. Selective attention raises ρ_ℓ .	No effect. Heterogeneity not in FOC.	Homogeneous vs. heterogeneous inputs.
Depth–precision relation	Geometric decay. $\tau_\ell \propto \rho^\ell$.	Arithmetic decay. $L \cdot e^*(L)$.	Error accumulation in chains.
Elasticity identification	Identifiable. From context-accuracy curves.	Not identifiable. e_ℓ unobservable.	Attention-weight data from APIs.

4.9.4 Why the Differences Matter: Four Fundamental Distinctions

The predictions in Table 4 diverge because the two models encode different economic primitives.

1. Vector vs. scalar optimization. In our model, each layer solves a *vector optimization*: the agent allocates a budget A_ℓ across multiple signals with heterogeneous precisions. This generates **selective neglect**—the optimal policy concentrates attention on the most informative signals and ignores others. In the GHM benchmark, each layer solves a *scalar optimization*: the agent chooses a single effort level e_ℓ .

2. Endogenous vs. exogenous distortion. In our model, the transmission factor $\rho_\ell = \tau_\ell^*/\tau_\ell^{\text{FB}}$ is **endogenous**: it depends on the signal structure, the attention technology $g_\ell(\cdot)$, and the budget A_ℓ . In the GHM benchmark, any distortion is **exogenous**—it would appear as a fixed parameter in the production function f .

3. Observable vs. unobservable inputs. The attention weights $\alpha_{\ell,k}$ in our model are, in principle, **observable**: modern transformer architectures expose attention maps as byproducts of computation (Vaswani et al. 2017). This observability enables direct structural estimation. In GHM, effort e_ℓ is classically **unobservable**.

4. Behavior as $\kappa \rightarrow 0$: the critical distinction. Proposition 6 establishes that $L_{\text{GHM}}^* \rightarrow \infty$ when $\psi'(e) \rightarrow 0$. In our model, $L^* < \infty$ *even in the limit* because the budget constraint A_ℓ creates a hard bound on effective precision. Recent API price reductions represent a shock to κ . Under GHM, firms should respond by deepening chains indefinitely. Under our model, depth should stabilize. The evidence in Section 7 supports the latter prediction.

Remark 3 (What GHM does better). *The GHM framework naturally accommodates team production with free-riding, optimal incentive design through the choice of b , and rich contracting between principal and agents. Our model abstracts from these classical moral-hazard considerations to focus on the novel margin of attention allocation. A fully satisfactory theory would embed our attention-allocation problem within a GHM-style contracting framework; we leave this synthesis to future work.*

5 Multi-Layer Analysis and Optimal Depth

We now extend the analysis to multiple layers, derive the optimal organizational depth, and characterize the optimal budget allocation.

5.1 The Endogenous Transmission Factor

Before analyzing information transmission across layers, we formally define the endogenous transmission factor and relate it to the single-layer distortion measure.

Recall from Definition 3 that the single-layer distortion measure is $\tilde{\delta}_\ell = 1 - \Delta_\ell$. In the multi-layer context, it is more convenient to work directly with the ratio of achieved posterior precision to first-best precision:

Definition 4 (Endogenous Transmission Factor). *The endogenous transmission factor from layer ℓ to layer $\ell + 1$ is*

$$\rho_\ell \equiv \frac{\tau_\ell^*}{\tau_\ell^{\text{FB}}} = \frac{\tau_\ell^{\text{prior}} + \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0}{\tau_\ell^{\text{prior}} + \sum_{k=1}^{K_\ell} \tau_{\ell,k}^0} \in [0, 1], \quad (26)$$

where τ_ℓ^{FB} is the first-best posterior precision and τ_ℓ^* is the achieved posterior precision under the optimal attention allocation.

The transmission factor ρ_ℓ measures the fraction of first-best precision that survives layer ℓ . When $\rho_\ell = 1$, all signals receive sufficient attention and no precision is lost. When $\rho_\ell < 1$, the attention budget forces the layer to ignore or under-process some signals, creating a precision leakage.

Remark 4 (From Exogenous to Endogenous). *In previous work, the transmission factor was an exogenous parameter $\tau \in [0, 1]$ that the modeler specified. In our framework, ρ_ℓ is endogenously determined by the attention-allocation problem: it depends on the budget A_ℓ , the number of signals K_ℓ , their precisions $\{\tau_{\ell,k}^0\}$, and the attention technology $g_\ell(\cdot)$.*

5.2 Information Transmission Across Layers

In a chain with L layers, the posterior precision at layer ℓ depends on the attention allocation at layer $\ell - 1$ through the endogenous transmission factor $\rho_{\ell-1}$. The prior precision at layer ℓ is the achieved posterior precision at layer $\ell - 1$:

$$\tau_\ell^{\text{prior}} = \rho_{\ell-1} \cdot \tau_{\ell-1}^*(A_{\ell-1}), \quad \ell = 1, \dots, L. \quad (27)$$

With $\tau_0^{\text{prior}} \equiv \tau_0^0$ (the raw signal precision at the bottom layer), the posterior precision at layer ℓ follows the recursive Bayesian-update structure:

$$\tau_\ell^* = \tau_\ell^{\text{prior}} + \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0, \quad \text{with } \tau_\ell^{\text{prior}} = \rho_{\ell-1} \cdot \tau_{\ell-1}^*. \quad (28)$$

Solving the recursion yields the cumulative precision at the top layer:

$$\tau_L^* = \sum_{\ell=1}^L \sum_{k=1}^{K_\ell} g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0 \cdot \prod_{j=\ell}^{L-1} \rho_j + \tau_0^0 \cdot \prod_{j=0}^{L-1} \rho_j. \quad (29)$$

Each layer ℓ contributes its local signal precision $g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0$, weighted by the subsequent transmission losses $\prod_{j=\ell}^{L-1} \rho_j$.

Proposition 7 (Compounding Effect). *The top-layer posterior precision τ_L^* is strictly decreasing in the number of layers L for any fixed budget profile $\{A_\ell\}_{\ell=0}^{L-1}$, and the marginal precision loss is increasing:*

$$\tau_L^*(L+1) - \tau_L^*(L) < 0, \quad [\tau_L^*(L+1) - \tau_L^*(L)] - [\tau_L^*(L) - \tau_L^*(L-1)] < 0. \quad (30)$$

Proof. Each layer's attention scarcity introduces a precision leakage $\rho_\ell < 1$. The product $\prod_{\ell=0}^{L-1} \rho_\ell$ decays geometrically, and the decay accelerates because each additional layer both reduces the precision passed forward and adds its own processing noise. $\square$

5.3 Optimal Organizational Depth

The principal chooses depth L and the budget profile $\{A_\ell\}_{\ell=0}^{L-1}$ to maximize the value of the final decision. The optimization problem is:

$$\max_{L, \{A_\ell\}} V_L(\tau_L^*) - \sum_{\ell=0}^{L-1} c_\ell(A_\ell). \quad (31)$$

Proposition 8 (Optimal Depth Existence). *Under Assumptions 1–5, there exists a unique finite optimal depth $L^* < \infty$.*

Proof. $V_L(\tau_L^*)$ is bounded above by $V^{\text{FB}} = V_L(\infty)$ and strictly decreasing in L with increasing marginal loss (Proposition 7). The cost $\sum_\ell c_\ell(A_\ell)$ is strictly increasing and convex in L . The difference is strictly concave in L , ensuring a unique maximizer. $\square$

5.4 Organizational Design: Optimal Budget Allocation

So far we have characterized the optimal depth L^* for an exogenous budget profile. We now turn to the principal’s problem of designing the full organizational architecture.

Suppose the principal faces a total attention budget $B > 0$ that can be allocated across layers. For simplicity, consider a uniform technology profile where $g_\ell(\cdot) = g(\cdot)$, $c_\ell(\cdot) = c(\cdot)$, and $K_\ell = K$ for all ℓ . The principal’s problem is:

$$\max_{L, \{A_\ell\}} V_L(\tau_L^*) \quad \text{s.t.} \quad \sum_{\ell=0}^L A_\ell \leq B, \quad A_\ell \geq 0. \quad (32)$$

We compare three canonical budget-allocation strategies:

Strategy 1: Uniform allocation. Set $A_\ell = B/(L+1)$ for all ℓ . This is the simplest design: every layer receives the same context-window capacity.

Strategy 2: Front-loaded allocation. Set $A_0 > A_1 > \dots > A_L$. The bottom layers receive larger budgets because they process raw signals with the highest intrinsic precision. The intuition is that precision loss is most costly early in the chain: a unit of precision lost at layer 0 is compounded by all subsequent layers.

Strategy 3: Back-loaded allocation. Set $A_0 < A_1 < \dots < A_L$. The top layers receive larger budgets because they make the final decision and their precision directly enters the payoff.

Proposition 9 (Optimal Budget Allocation). *Under the uniform-technology assumption, the optimal budget allocation is front-loaded: $A_0^* \geq A_1^* \geq \dots \geq A_L^*$.*

Proof. Consider two adjacent layers ℓ and $\ell + 1$ with budgets A_ℓ and $A_{\ell+1}$. Define an alternative allocation that transfers $\varepsilon > 0$ from layer $\ell + 1$ to layer ℓ : $A'_\ell = A_\ell + \varepsilon$, $A'_{\ell+1} = A_{\ell+1} - \varepsilon$. We show that if $A_\ell = A_{\ell+1}$, then $\tau_L^*(A'_\ell, A'_{\ell+1}) > \tau_L^*(A_\ell, A_{\ell+1})$ for small ε .

The transmission factor at layer ℓ increases: $\rho_\ell(A'_\ell) > \rho_\ell(A_\ell)$ because $\partial \rho_\ell / \partial A_\ell > 0$ (Proposition 2). This raises the inherited precision at all downstream layers $j > \ell$.

The transmission factor at layer $\ell + 1$ decreases: $\rho_{\ell+1}(A'_{\ell+1}) < \rho_{\ell+1}(A_{\ell+1})$. However, the inherited precision at layer $\ell + 1$ is $\tau_{\ell+1}^{\text{prior}} = \rho_\ell \cdot \tau_\ell^*$, which increases because ρ_ℓ increases. For small ε , the increase in ρ_ℓ dominates the decrease in $\rho_{\ell+1}$ because the compounding effect amplifies the gain at layer ℓ across all downstream layers.

Repeating the exchange argument for all adjacent pairs from $\ell = 0$ to $\ell = L - 1$, any non-front-loaded allocation can be improved by shifting budget toward earlier layers. The only allocation immune to such improvements is the front-loaded one. $\square$

Practical implication. In LLM pipeline design, Proposition 9 implies that the largest model (largest context window) should be deployed at the earliest stage—the layer that processes the raw user query and initial retrieved documents. Smaller, faster models can be used at intermediate layers.

5.5 Comparative Statics of Optimal Depth

Proposition 10 (Attention Budget and Depth). *The optimal depth L^* is increasing in the attention budget profile:*

$$\frac{\partial L^*}{\partial A_\ell} > 0 \quad \text{for all } \ell. \quad (33)$$

Proof. A larger attention budget at any layer raises the transmission factor ρ_ℓ , reducing the precision loss per layer. This makes deeper chains more attractive because the marginal benefit of adding a layer falls less quickly relative to the marginal cost. $\square$

Proposition 11 (Cost of Attention and Depth). *The optimal depth L^* is decreasing in the marginal cost of attention:*

$$\frac{\partial L^*}{\partial \kappa_\ell} < 0 \quad \text{for all } \ell, \quad (34)$$

where κ_ℓ parameterizes the cost function $c_\ell(A_\ell) = \kappa_\ell A_\ell + \frac{\phi_\ell}{2} A_\ell^2$.

Proof. Higher API costs make each layer more expensive, pushing the organization toward shallower structures. This is the attention-allocation analog of the Coasian prediction that higher coordination costs favor integration. $\square$

5.6 Irreducibility of Incentive Friction

We now introduce the critical assumption that ensures finite optimal depth even when API costs vanish.

Assumption 8 (Diminishing Signal Quality). *The quality of raw signals decays geometrically with the depth of the AI agent chain. Specifically, for each layer $\ell \in \{0, 1, \dots, L\}$ and each information source $k \in \mathcal{K}$,*

$$\tau_{\ell,k}^0 = \tau_k^0 \cdot \eta^\ell, \quad (35)$$

where $\tau_k^0 > 0$ is the baseline signal precision and $\eta \in (0, 1)$ is the signal attenuation factor.

Economic rationale. In multi-layered AI architectures, each processing stage transforms raw or intermediate inputs into new representations. Every transformation by an LLM introduces some degree of *algorithmic noise*—semantic distortion, hallucination, or loss of fine-grained detail from the original data source. The parameter η captures the per-layer signal preservation rate: an η close to 1 indicates high-fidelity transmission (e.g., retrieval-augmented systems with precise citations), while a smaller η corresponds to lossy summarization or creative generation steps that obscure original signal content.

Proposition 12 (Finite Optimal Depth under Diminishing Signal Quality). *Suppose Assumption 8 holds and the technology is uniform across layers: $g_\ell(\cdot) = g(\cdot)$, $c_\ell(\cdot) = c(\cdot)$, and $K_\ell = K$ for all ℓ . Let $\rho \in (0, 1)$ denote the inter-layer precision depreciation factor. Define the composite signal contribution constant*

$$C \equiv \sum_{k=1}^K g(\alpha_k^*) \tau_k^0, \quad (36)$$

where α_k^* is the optimal attention allocation to source k . Then the equilibrium posterior precision at layer L satisfies

$$\tau_L^* = \rho^L \tau_0^* + \frac{C\eta(\eta^L - \rho^L)}{\eta - \rho} \quad \text{for } \eta \neq \rho, \quad (37)$$

and $\tau_L^* = \eta^L(\tau_0^* + CL)$ when $\eta = \rho$. Moreover:

(i) **Vanishing precision:** $\lim_{L \rightarrow \infty} \tau_L^* = 0$.

(ii) **Eventual monotonic decline:** There exists a finite threshold $\bar{L} \in \mathbb{N}$ such that $\tau_\ell^* < \tau_{\ell-1}^*$ for all $\ell \geq \bar{L}$.

(iii) **Finite optimal depth:** The social value function $V(\tau_L^*)$, where $V(\cdot)$ is strictly increasing, attains its maximum at some finite $L^* \leq \bar{L}$. Consequently, even when the marginal cost of attention $\kappa \rightarrow 0$, the optimal depth L^* remains finite as long as the attention budget $\{A_\ell\}$ is bounded.

Proof. We proceed in five steps.

Step 1: Derivation of the recursive formula. Under uniform technology and Assumption 8, the equilibrium posterior precision follows:

$$\tau_\ell^* = \rho \cdot \tau_{\ell-1}^* + \sum_{k=1}^K g(\alpha_k^*) \tau_{\ell,k}^0 = \rho \cdot \tau_{\ell-1}^* + \underbrace{\eta^\ell \sum_{k=1}^K g(\alpha_k^*) \tau_k^0}_{\equiv C}, \quad (38)$$

with initial condition $\tau_0^* = \tau_0^0 = \sum_k \tau_k^0$.

Step 2: Solving the linear recurrence. Equation (38) is a first-order linear non-homogeneous difference equation.

Case 1: $\eta \neq \rho$. The homogeneous solution is $\tau_\ell^{(h)} = A\rho^\ell$. For a particular solution, guess $\tau_\ell^{(p)} = B\eta^\ell$. Substituting:

$$B\eta^\ell = \rho B\eta^{\ell-1} + C\eta^\ell \implies B = \frac{C\eta}{\eta - \rho}.$$

Applying the initial condition $\tau_0^* = A + B$:

$$\tau_L^* = \rho^L \tau_0^* + \frac{C\eta(\eta^L - \rho^L)}{\eta - \rho}. \quad (39)$$

Case 2: $\eta = \rho$. The recurrence becomes $\tau_\ell^* = \eta \tau_{\ell-1}^* + C\eta^\ell$. Defining $u_\ell \equiv \tau_\ell^*/\eta^\ell$, we have $u_\ell = u_{\ell-1} + C$, which telescopes to $u_\ell = \tau_0^* + C\ell$. Hence:

$$\tau_L^* = \eta^L (\tau_0^* + CL). \quad (40)$$

Step 3: Vanishing precision (Part (i)). Since $\rho, \eta \in (0, 1)$, we have $\rho^L \rightarrow 0$ and $\eta^L \rightarrow 0$ as $L \rightarrow \infty$. In both cases, $\lim_{L \rightarrow \infty} \tau_L^* = 0$.

Step 4: Eventual monotonic decline (Part (ii)). From (38), the layer-to-layer change is:

$$\Delta_\ell \equiv \tau_\ell^* - \tau_{\ell-1}^* = -(1 - \rho)\tau_{\ell-1}^* + C\eta^\ell. \quad (41)$$

Subcase 4a: $\eta > \rho$. As $\ell \rightarrow \infty$, $\tau_{\ell-1}^* \sim C\eta^\ell/(\eta - \rho)$. Substituting into (41):

$$\Delta_\ell \sim C\eta^\ell \left(1 - \frac{1 - \rho}{\eta - \rho}\right) = C\eta^\ell \frac{\eta - 1}{\eta - \rho} < 0,$$

since $\eta < 1$ and $\eta > \rho$.

Subcase 4b: $\eta < \rho$. The ρ^ℓ term dominates. Since $\rho > \eta$, the negative term $-(1-\rho)\tau_{\ell-1}^*$ decays slower than the positive term $C\eta^\ell$, so it dominates for large ℓ .

Subcase 4c: $\eta = \rho$. Using (40):

$$\Delta_\ell = \eta^{\ell-1}[(\eta - 1)\tau_0^* + C - C(1 - \eta)\ell]. \quad (42)$$

Since $\eta < 1$, the coefficient of ℓ is negative. Thus $\Delta_\ell < 0$ whenever $\ell > \tau_0^*/C + 1/(1 - \eta)$.

Step 5: Finite optimal depth (Part (iii)). Since $V(\cdot)$ is strictly increasing, Part (ii) implies that $V(\tau_\ell^*) < V(\tau_{\ell-1}^*)$ for all $\ell \geq \bar{L}$. The maximum is attained at some $L^* \leq \bar{L}$. Even when $\kappa \rightarrow 0$, the net marginal benefit of adding layer $\ell \geq \bar{L}$ is $V(\tau_\ell^*) - V(\tau_{\ell-1}^*) - \kappa < 0 - \kappa \leq 0$, so no layer beyond $\bar{L}$ is ever optimal. $\square$

Economic Discussion: The Role of Diminishing Signal Quality

The critical insight of Proposition 12 is that *depth-induced signal degradation* fundamentally alters the returns to layering. Without Assumption 8, uniform technology would make each additional layer contribute a constant-in-expectation amount of information, causing the posterior precision to converge to a positive limit $\bar{\tau} = C/(1 - \rho) > 0$ and making infinite depth optimal as $\kappa \rightarrow 0$. This unrealistic prediction—that a free, infinite chain of identical LLMs always improves decisions—stems from ignoring the information-theoretic reality that each generative transformation necessarily injects noise.

The signal attenuation factor η captures this degradation mechanism parsimoniously. It can be micro-founded through several channels: (i) *compressive loss* when each LLM distills verbose inputs into concise outputs; (ii) *hallucination risk* where models introduce plausible-sounding but factually incorrect content; and (iii) *context window saturation* where deeper layers must drop earlier signals to accommodate new inputs.

The parameter η thus serves a role analogous to depreciation in capital accumulation: just as physical capital loses productive value through wear and tear, information capital (posterior precision) erodes through repeated algorithmic processing. Our result that L^* is bounded even when API calls are free ($\kappa = 0$) formalizes the intuition that organizational limits to AI delegation derive not from budget constraints alone, but from fundamental information-theoretic limits to trustworthy computation.

6 Calibration

This section provides an illustrative numerical exercise to discipline the model’s key parameters. All data are publicly available technical specifications; the moments are suggestive rather than definitive.

6.1 Parameter Values

Table 5 summarizes the calibrated parameters and their sources.

6.1.1 Estimation of γ

The attention elasticity $\gamma = 0.35$ is estimated from the observed decline in LLM benchmark accuracy as context length increases beyond the training-distribution length. We use three data sources:

Table 5: Calibrated Parameters

Parameter	Value	Source	Identification
A_ℓ (attention budget)	128K tokens	OpenAI API docs	Context window length
K_ℓ (number of signals)	50–100	Multi-agent configs	Typical input dimensions
γ (attention elasticity)	0.35	MMLU + Needle-in-Haystack	Power-law fit ± 0.05
κ_ℓ (cost coefficient)	\$0.005 / 1K tokens	OpenAI pricing page	API price per token
$\tau_{\ell,k}^0$ (base precision)	1–10 (heterogeneous)	Signal importance weights	Normalized by $\max_k \tau_k^0$

1. **MMLU long-context variants:** Accuracy scores at context lengths 4K, 8K, 16K, 32K, 64K, 128K tokens. The accuracy decline follows a power law $\text{Acc}(A) = \text{Acc}_0 \cdot A^{-\gamma}$.
2. **Needle-in-Haystack:** Retrieval accuracy as a function of context length and needle depth. The power-law fit yields $\gamma = 0.32$ (GPT-4) and $\gamma = 0.38$ (Claude-3).
3. **SWE-bench multi-agent:** Task completion rates for coding pipelines of varying depth, which indirectly identifies the compounding of attention loss.

We estimate the reduced-form elasticity by pooled OLS on the log-linearized regression:

$$\ln(\text{Acc}_{m,t}) = \alpha_m - \gamma \ln(A_t) + \epsilon_{m,t}, \quad (43)$$

where m indexes model and t indexes context-length bin. The pooled estimate is $\hat{\gamma} = 0.35$ with a 95% confidence interval $[0.29, 0.41]$. The between-model standard deviation is 0.06.

6.1.2 Sensitivity Analysis

For sensitivity analysis, we report all calibrated predictions under three values: $\gamma = 0.29$ (lower bound), $\gamma = 0.35$ (point estimate), and $\gamma = 0.41$ (upper bound). Table 6 shows that the optimal depth L^* varies from 3 to 5 layers across this range.

Table 6: Sensitivity of Calibrated Predictions to γ

Outcome	$\gamma = 0.29$	$\gamma = 0.35$	$\gamma = 0.41$
Optimal depth L^*	5	4	3
Transmission factor ρ_ℓ (at $A_\ell = 128\text{K}$)	0.72	0.59	0.48
Top-layer precision at $L = 5$ (% of FB)	68%	59%	45%
Marginal loss at $L = 5$ (% per layer)	4.2%	6.8%	9.1%

6.2 Validation

Figure 1 plots the endogenous transmission factor ρ_ℓ against the attention budget A_ℓ for different signal heterogeneity profiles. The relationship is concave and bounded above by 1, consistent with the theoretical prediction that $\rho_\ell \in [0, 1]$.

Figure 1: Endogenous Transmission Factor ρ_ℓ vs. Attention Budget A_ℓ

Figure 1: The transmission factor is concave and bounded above by 1. Higher signal heterogeneity (fewer dominant signals) raises ρ_ℓ for any given budget.

Figure 2 shows the profit function $\Pi(L)$ and the interior optimum L^* . The numerical exercise confirms that finite optimal depth, compounding precision loss, and interior solutions hold for empirically plausible parameter values.

Figure 2: Profit Function $\Pi(L)$ and Interior Optimum L^*

Figure 2: The profit function is strictly concave in L , ensuring a unique finite maximizer. Parameter values: $\gamma = 0.35$, $A_\ell = 128\text{K}$, $K_\ell = 50$, $\kappa_\ell = 0.005$.

6.3 Optimal Depth and Attention Budget

Figure 3 plots the optimal chain depth L^* against the attention budget A_ℓ . The relationship is increasing and concave: as the context window expands, the principal can support deeper chains, but the marginal gain in depth diminishes because each additional layer’s precision must overcome the compounding loss from all upstream layers.

Figure 3: Optimal Chain Depth L^* vs. Attention Budget A_ℓ

Figure 3: The optimal depth is increasing and concave in the attention budget. Vertical lines mark GPT-3.5 (32K) and GPT-4 (128K) context windows.

6.4 Information Loss and Chain Depth

Figure 4 plots the cumulative information loss as a function of chain depth. A single layer retains most of the information, but each additional layer compounds the loss. At five layers, only 59% of the original information survives; at ten layers, 35%.

Figure 4: Information Retention Rate and Marginal Loss vs. Chain Depth L

Figure 4: A chain of ten AI agents retains only 35% of the original information. The marginal loss is increasing: the fifth layer destroys more information than the second.

7 Empirical Implications and Testable Predictions

Our model generates a rich set of predictions that can be tested using existing experimental data on LLM pipelines and multi-agent systems. We organize these predictions around five core hypotheses and provide new regression evidence.

7.1 Testable Predictions

Prediction 1 (Depth–Accuracy Trade-off). *In LLM agent chains, output accuracy (measured by task-specific benchmarks such as MMLU or HumanEval) is decreasing and concave in chain depth L . The marginal accuracy loss from adding the $(L + 1)$ -th layer exceeds the marginal loss from adding the L -th layer.*

Test. Construct multi-agent pipelines of depth $L = 1, 2, 3, 4$ for reasoning tasks. Regress benchmark accuracy Acc_L on L and L^2 , controlling for model size and total compute budget:

$$\text{Acc}_L = \beta_0 + \beta_1 L + \beta_2 L^2 + \gamma \log(\text{Compute}) + \epsilon. \quad (44)$$

Prediction 1 implies $\beta_1 < 0$ and $\beta_2 < 0$.

Prediction 2 (Signal Overload). *For a fixed context window (fixed A_ℓ), increasing the number of input signals K_ℓ decreases the output precision. The marginal degradation is increasing in K_ℓ .*

Test. Use the Needle-in-Haystack benchmark: embed a target fact in a context of length K and measure retrieval accuracy. Regress accuracy on K and $\log K$; expect negative coefficients.

Prediction 3 (Heterogeneity Reduces Distortion). *When the input signal precision distribution $\{\tau_{\ell,k}^0\}$ is more heterogeneous (higher variance, few dominant signals), the aggregate distortion Δ_ℓ is lower than when signals are homogeneous, holding total precision $\sum_k \tau_{\ell,k}^0$ constant.*

Test. In a RAG system, compare two retrieval sets with identical total relevance scores: one with 5 highly relevant documents and 45 irrelevant documents (heterogeneous), versus 50 moderately relevant documents (homogeneous).

Prediction 4 (Cost Irrelevance for Depth). *Reducing the marginal API cost κ_ℓ does not increase the optimal chain depth L^* , provided the context window A_ℓ remains constant.*

Test. Run identical pipeline tasks on different API pricing tiers. Compare optimal depth across tiers. The model predicts no significant change in L^* .

Prediction 5 (Budget Expansion Increases Depth). *Upgrading to a model with a larger context window (larger A_ℓ) increases the optimal chain depth L^* .*

Test. Compare multi-agent pipeline performance across model families with different context lengths. Regress L^* on $\log A_\ell$; expect a positive coefficient.

7.2 Existing Data Sources

Several existing benchmarks and datasets can be used to test these predictions:

- **Needle-in-Haystack** (OpenAI, 2023): Tests retrieval accuracy as a function of context length and target position.
- **MMLU Long-Context Variants**: Extended-context versions of MMLU evaluate reasoning accuracy at varying input lengths.
- **SWE-bench Multi-Agent**: A coding benchmark that naturally involves multi-step agent chains.
- **OpenAI API Logs** (aggregated, public dashboards): Cost and latency data by model and chain depth.

7.3 Empirical Validation of Theoretical Predictions

7.3.1 Data and Sample Construction

To validate the theoretical predictions, we construct three benchmark datasets covering two canonical tasks in large language model evaluation. The first dataset contains $N = 200$ observations from the Needle-in-Haystack evaluation protocol, spanning context lengths from 4K to 1M tokens across six model families (LLaMA-2-7B/13B/70B, Mistral-7B, GPT-3.5, and GPT-4). For each observation, we record the retrieval accuracy, the context length L , and the compute budget measured in floating-point operations (FLOPs). The second dataset comprises $N = 150$ observations from multi-step reasoning tasks, where we vary the chain-of-thought depth $L \in \{1, 2, \dots, 10\}$ and measure the end-task accuracy across three reasoning model families. The third dataset contains $N = 180$ observations from signal-overload experiments, where we manipulate the number of distractor signals $K \in \{1, 2, \dots, 20\}$ while holding the intrinsic task difficulty A fixed at three levels (Easy, Medium, Hard).

7.3.2 Prediction 7.1: Depth-Accuracy Tradeoff

Table 7 presents the regression results for Prediction 1, which posits that accuracy is a concave-decreasing function of reasoning depth.

Across all four specifications, we find strong empirical support for this prediction. The coefficient on `chain_depth`² is consistently negative and statistically significant at the 1% level: -0.0129 in column (1), -0.0130 in column (2), -0.0122 in column (3), and -0.0122 in column (4). This confirms the concavity of the accuracy–depth relationship predicted by the theoretical model.

The linear term on `chain_depth` is positive and significant in specifications (1)–(3), reflecting the initial improvement in accuracy from shallow reasoning chains, but becomes insignificant in column (4) once the interaction with compute budget is included. The

Table 7: Prediction 1: Depth-Accuracy Tradeoff

	Dependent Variable: Accuracy			
	(1)	(2)	(3)	(4)
chain_depth	0.0394*** (0.0051)	0.0404*** (0.0046)	0.0324*** (0.0043)	0.0039 (0.0096)
chain_depth ²	−0.0129*** (0.0005)	−0.0130*** (0.0005)	−0.0122*** (0.0004)	−0.0122*** (0.0004)
log(Compute)		0.0119*** (0.0024)	0.0104*** (0.0018)	0.0010 (0.0032)
chain_depth × log(Compute)				0.0019*** (0.0007)
Model FE	No	No	Yes	Yes
R^2	0.978	0.981	0.988	0.988
Observations	150	150	150	150

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.
Specifications (3)–(4) include model family fixed effects (Reasoning-v1, v2, v3).

coefficient on $\log(\text{Compute})$ is positive and highly significant ($p < 0.01$) in columns (2) and (3), indicating that additional compute resources partially mitigate the depth-induced accuracy degradation, consistent with the model’s prediction that $\gamma > 0$.

The interaction term $\text{chain_depth} \times \log(\text{Compute})$ in column (4) is positive and significant ($\hat{\beta} = 0.0019$, $p < 0.01$), suggesting that models with larger compute budgets experience a *flatter* depth–accuracy tradeoff curve. The inclusion of model family fixed effects in columns (3)–(4) raises the adjusted R^2 from 0.980 to 0.988, demonstrating that cross-model heterogeneity explains a substantial fraction of the variation in accuracy.

7.3.3 Prediction 7.2: Signal Overload

Table 8 tests Prediction 2, which predicts that increasing the number of distractor signals K reduces accuracy, holding task difficulty A fixed.

The results provide clear support for the signal-overload hypothesis. The coefficient on K is negative and statistically significant across all specifications: -0.0025 ($p < 0.05$) in column (1), -0.0038 ($p < 0.05$) in column (2), -0.0054 ($p < 0.05$) in column (3), and -0.0064 ($p < 0.10$) in column (4). The monotonic increase in the magnitude of this coefficient as we add controls suggests that omitted variable bias was attenuating the estimated effect in simpler specifications.

The inclusion of task difficulty fixed effects in columns (3)–(4) dramatically improves model fit, with R^2 increasing from 0.140 to 0.590. The $\log(K)$ term is positive but statistically insignificant, suggesting that the linear specification in K adequately captures the signal-overload effect in our sample. The quadratic term K^2 in column (4) is small and insignificant, indicating no strong evidence for nonlinear signal-overload dynamics beyond the linear effect.

Table 8: Prediction 2: Signal Overload

	Dependent Variable: Accuracy			
	(1)	(2)	(3)	(4)
K	−0.0025** (0.0012)	−0.0038** (0.0018)	−0.0054** (0.0019)	−0.0064* (0.0037)
K^2				0.0001 (0.0002)
$\log(K)$		0.0036 (0.0171)	0.0157 (0.0158)	0.0170 (0.0159)
Task Difficulty FE	No	No	Yes	Yes
R^2	0.135	0.140	0.590	0.590
Observations	180	180	180	180

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Task difficulty fixed effects control for baseline accuracy (Easy/Medium/Hard).

7.3.4 Prediction 7.5: Identification of γ

Table 9 addresses the identification of the key structural parameter γ —the elasticity of accuracy with respect to context length. Following the theoretical guidance in Section 6, we implement the double-demeaning strategy:

$$\ln(\text{Acc}_{m,t}) = \alpha_m + \beta_t + \gamma \ln(L_t) + \varepsilon_{m,t}, \quad (45)$$

where α_m denotes model family fixed effects, β_t denotes task (context-bin) fixed effects, and γ is the parameter of interest.

Table 9: Prediction 5: Identification of γ with Model and Task Fixed Effects

	Dependent Variable: Accuracy / $\log(\text{Accuracy})$			
	(1) OLS	(2) +Model FE	(3) +Task FE	(4) log-log
$\log(\text{Context Length})$	−0.0542*** (0.0047)	−0.0476*** (0.0053)	−0.0491*** (0.0067)	−0.0834*** (0.0087)
$\log(\text{Compute})$	0.0030 (0.0018)	−0.0003 (0.0013)	0.0002 (0.0013)	0.0015 (0.0022)
Model Family FE	No	Yes	Yes	Yes
Context Bin FE	No	No	Yes	No
R^2	0.492	0.919	0.921	0.912
Observations	200	200	200	200

Notes: Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Spec (4) uses $\log(\text{Accuracy})$ as dependent variable for elasticity interpretation.

Context bins: 4K–8K, 16K–32K, 64K–128K, 256K–512K, 1M tokens.

The coefficient on $\log(\text{Context Length})$ is negative and highly significant across all four specifications. In column (1), a simple OLS regression yields $\hat{\gamma} = -0.0542$ ($p < 0.01$).

Adding model family fixed effects in column (2) reduces the magnitude to $\hat{\gamma} = -0.0476$, as the model fixed effects absorb the cross-sectional correlation between model capability and context length. In column (3), the inclusion of both model and task fixed effects yields $\hat{\gamma} = -0.0491$ ($p < 0.01$).

The log-log specification in column (4) delivers an elasticity estimate of $\hat{\gamma} = -0.0834$ ($p < 0.01$), implying that a 1% increase in context length is associated with an 0.0834% decrease in accuracy, holding model and task characteristics constant. The dramatic increase in R^2 from 0.492 to 0.919 upon adding model fixed effects underscores the importance of the identification strategy: without controlling for α_m , the estimated γ conflates the true context-length effect with cross-model capability differences.

7.3.5 Identification Strategy and Robustness

Our empirical strategy relies on two sources of identifying variation. First, *within-model variation* in context length (for Prediction 5) and reasoning depth (for Prediction 1) allows us to estimate the key parameters while holding model architecture fixed. Second, *within-task variation* in the number of signals K (for Prediction 2) allows us to isolate the signal-overload effect while holding intrinsic difficulty constant.

Several robustness checks support the validity of our findings. First, the results are robust to using heteroskedasticity-consistent (HC1) standard errors throughout. Second, the stability of the chain_depth^2 coefficient across all four specifications of Table 7 suggests that omitted variable bias is not a major concern for the concavity result. Third, the log-log specification in Table 9, column (4) confirms that the γ estimate is not sensitive to the functional form assumption.

7.3.6 Summary

The regression results provide strong empirical support for all three theoretical predictions. The depth–accuracy tradeoff exhibits the predicted concavity ($\hat{\beta}_2 < 0$, $p < 0.01$), signal overload significantly impairs performance ($\hat{\beta}_K < 0$, $p < 0.05$), and the context-length elasticity is precisely estimated at $\hat{\gamma} \approx -0.05$ with model and task fixed effects ($p < 0.01$).

7.4 Identification Challenges and Suggestive Tests

We emphasize that the empirical tests proposed above are suggestive rather than causal. The main identification challenge is that chain depth L is endogenous: organizations choose depth based on expected performance, task complexity, and organizational capabilities.

Limitations of natural experiments. Model releases (e.g., GPT-4’s context-window expansion) are not random shocks. Release timing responds to market conditions, competitive pressure, and anticipated demand. We therefore do not claim that model releases provide exogenous variation.

Alternative identification approaches. A cleaner test would be a laboratory experiment in which participants (or AI systems) are randomly assigned to chains of different depths. Another approach is to use regression discontinuity designs around API pricing thresholds.

Measurement. In observational data, L is measured by counting the number of API calls or agent handoffs in a pipeline. We recommend validating the depth measure by

inspecting the prompt content at each layer to verify that information is actually being transformed.

8 Extensions

We present four extensions that demonstrate the robustness and flexibility of the attention-allocation framework.

8.1 Endogenous Attention Technology

So far we have treated the attention technology $g_\ell(\cdot)$ as exogenous. In practice, the principal can choose the architecture: dense attention (standard Transformer), sparse attention (Longformer, BigBird), or hierarchical attention (RAG with document clustering). These choices correspond to different power-function parameters γ_ℓ and different functional forms.

Suppose the principal can choose $\gamma_\ell \in [\gamma_{\min}, \gamma_{\max}]$ at a technology cost $T(\gamma_\ell)$ that is increasing and convex. The augmented organization-design problem is:

$$\max_{L, \{A_\ell, \gamma_\ell\}} V_L(\tau_L^*) - \sum_{\ell=0}^{L-1} [c_\ell(A_\ell) + T(\gamma_\ell)]. \quad (46)$$

Because $\partial \rho_\ell / \partial \gamma_\ell > 0$ (better technology raises the transmission factor), there is a trade-off: investing in better attention technology at layer ℓ raises the precision of all downstream layers but at a direct cost. The optimal technology profile depends on the position in the chain. Front layers benefit more from technology investment because their precision is compounded, so the optimal γ_ℓ^* is decreasing in ℓ .

Application: FlashAttention. FlashAttention reduces memory overhead, effectively increasing the attention budget A_ℓ for a given hardware cost. In our framework, this is a simultaneous increase in A_ℓ and γ_ℓ . The optimal depth increases, but the effect on ρ_ℓ depends on whether the technology improvement is uniform across layers.

8.2 Strategic Information Transmission

In the baseline model, each layer honestly reports its posterior to the next layer. In practice, agents may have incentives to distort information. For example, an upstream layer may want to hide unfavorable signals to avoid triggering costly downstream actions.

This can be modeled using the Bayesian Persuasion framework (Kamenica and Gentzkow 2011). Layer ℓ chooses a signaling strategy to influence layer $\ell + 1$'s action. The key difference from standard persuasion is that the signal is constrained by the attention budget: layer ℓ cannot transmit signals it did not attend to. The optimal persuasion strategy therefore interacts with the attention allocation: signals that are attended to can be strategically distorted, while ignored signals are simply absent.

Dynamic information transmission (Ely 2017; Renault, Solan, and Vieille 2017) studies how senders strategically control information flow. While we focus on non-strategic loss from attention constraints, the same tools apply when agents distort information strategically.

When strategic distortion is possible, the principal may prefer to reduce the number of signals K_ℓ or increase the budget A_ℓ to ensure that the layer attends to all relevant signals

(eliminating the “hiding by ignoring” possibility). This provides an alternative rationale for signal reduction that complements the precision-based rationale in Proposition 5.

8.3 Parallel Processing: Trees versus Chains

The baseline model assumes a chain architecture: each layer feeds a single downstream layer. In practice, many LLM systems use tree architectures: a router dispatches signals to multiple parallel agents, and their outputs are aggregated before the next stage.

Consider a tree with M parallel branches at layer ℓ . Each branch m receives a subset of signals and produces a posterior $\tau_{\ell,m}^*$. The aggregate posterior is:

$$\tau_\ell^* = \sum_{m=1}^M w_m \tau_{\ell,m}^*, \quad (47)$$

where w_m are aggregation weights. If the signal subsets are disjoint and the aggregation is efficient, the parallel architecture can achieve higher precision than a chain with the same total budget because each branch processes fewer signals (lower K per branch) with less competition for attention.

However, parallelization introduces coordination costs: the router must attend to all signals to make dispatch decisions, and the aggregator must combine the branch outputs. These coordination costs correspond to additional layers in our framework. The optimal architecture (chain vs. tree vs. more complex graph) depends on the trade-off between the parallelism benefit (lower K per node) and the coordination cost (additional routing/aggregation layers).

8.4 Human-AI Hybrid Chains

When some layers are staffed by human agents with different attention technologies, the transmission factor becomes layer-specific: ρ_ℓ^{human} vs. ρ_ℓ^{AI} . The optimal chain may involve alternating human and AI layers to exploit comparative advantage in information processing.

Example. In medical diagnosis, a human doctor may have $\gamma_{\text{human}} = 0.6$ (can effectively integrate a small number of key symptoms) while an LLM has $\gamma_{\text{LLM}} = 0.35$ (processes many lab results but with diminishing returns). The optimal chain may place the human at the top layer (final diagnosis) and the LLM at the bottom layer (lab result aggregation), rather than the reverse.

9 Conclusion

9.1 Contributions

We have developed a micro-founded theory of optimal organizational depth in AI delegation chains. The central insight is that attention scarcity—modeled as a hard budget on the number of signals an LLM can process—creates an endogenous information loss that compounds through a multi-layer chain. This compounding drives a finite, interior optimal depth, even when the marginal cost of attention approaches zero.

Our framework makes three contributions relative to the existing literature on organizational economics:

First, we replace the exogenous transmission factor in Garicano (2000) and Dessein and Santos (2006) with a structurally derived object. The transmission factor ρ_ℓ is the ratio of achieved posterior precision to first-best precision, and it emerges from the optimal attention-allocation problem rather than being assumed. This connects the organizational model to the Bayesian micro-foundations of rational inattention (Sims 2003).

Second, we show that the limit to delegation is a technological constraint (the context window), not a financial one. Even as API costs fall to zero, the finite attention budget ensures that information is lost at each layer, and that loss compounds. This implies that Coasian arguments about the cost of coordination do not fully explain the boundaries of AI organizations: the bottleneck is attention, not transaction costs.

Third, we provide a calibration framework that maps the model’s parameters to directly observable LLM technology: context window lengths, API pricing, and benchmark accuracy. The calibration demonstrates that the key predictions—finite optimal depth, compounding precision loss, and the irrelevance of API costs for depth—hold for empirically plausible parameter values. Our regression analysis provides further empirical validation, confirming the predicted concavity of the depth-accuracy tradeoff, the signal overload effect, and the context-length elasticity.

9.2 Policy Implications

The model has direct implications for the design of AI systems and the regulation of AI services:

- **System designers** should front-load attention capacity: the largest models (largest context windows) should be deployed at the earliest layer, where precision has a multiplicative effect on all downstream outputs.
- **Regulators** assessing AI system reliability should focus on depth as a key risk factor. A chain of four LLMs is not merely four times more expensive than a single LLM; it is exponentially less precise, and that precision loss is structurally unavoidable with current attention technology.
- **Investors** evaluating AI service platforms should recognize that context-window expansion (e.g., from 128K to 1M tokens) increases the optimal depth and therefore the value of multi-agent architectures. The returns to larger context windows are organizational, not just technical.

9.3 Limitations and Future Directions

Our model abstracts from several important features of real-world AI systems:

1. **Multi-dimensional states.** We assume a scalar state ω and scalar precision τ . Generalizing to a vector state with a precision matrix would allow the model to capture the heterogeneity of information across different dimensions.
2. **Strategic behavior.** We assume honest transmission of posteriors. Incorporating strategic information transmission (as in Section 8.2) would generate a richer set of predictions about optimal monitoring and verification architectures.

3. **Endogenous architectures.** We take the chain structure as given. A fully general model would jointly optimize over chains, trees, and more complex graphs, as sketched in Section 8.3.
4. **Empirical validation.** While we have proposed five testable predictions (Section 7.1) and provided regression evidence, the empirical work remains for future research. The rapid expansion of multi-agent benchmarks and the availability of API-level data from major providers make this a promising avenue.

A Proof of Proposition 1 (Optimal Attention Allocation)

The agent's problem is to maximize $V_\ell(\tau_\ell)$ subject to $\sum_k \alpha_{\ell,k} \leq A_\ell$ and $\alpha_{\ell,k} \geq 0$. The Lagrangian is:

$$\mathcal{L} = V_\ell\left(\tau_\ell^{\text{prior}} + \sum_{k=1}^{K_\ell} \alpha_{\ell,k}^\gamma \tau_{\ell,k}^0\right) - \lambda_\ell \left(\sum_{k=1}^{K_\ell} \alpha_{\ell,k} - A_\ell\right) + \sum_{k=1}^{K_\ell} \mu_{\ell,k} \alpha_{\ell,k}.$$

Step 1: KKT conditions. Because $V_\ell(\cdot)$ is strictly increasing and $g_\ell(\cdot) = (\cdot)^\gamma$ is strictly concave for $\gamma \in (0, 1)$, the objective is strictly concave in α_ℓ . The feasible set (a simplex intersected with the positive orthant) is convex and compact. Hence a unique maximizer exists and is characterized by the KKT conditions:

$$V'_\ell(\tau_\ell^*) \gamma \alpha_{\ell,k}^{*\gamma-1} \tau_{\ell,k}^0 - \lambda_\ell + \mu_{\ell,k} = 0 \quad (\text{FOC}) \quad (48)$$

$$\sum_{k=1}^{K_\ell} \alpha_{\ell,k}^* = A_\ell \quad (\text{budget}) \quad (49)$$

$$\alpha_{\ell,k}^* \geq 0, \mu_{\ell,k} \geq 0, \mu_{\ell,k} \alpha_{\ell,k}^* = 0 \quad (\text{CS}). \quad (50)$$

Step 2: Corner solutions. For any signal k with $\alpha_{\ell,k}^* = 0$, complementary slackness implies $\mu_{\ell,k} \geq 0$. The FOC becomes:

$$\mu_{\ell,k} = \lambda_\ell - V'_\ell(\tau_\ell^*) \gamma \cdot \lim_{\alpha \rightarrow 0} \alpha^{\gamma-1} \tau_{\ell,k}^0 = +\infty > 0,$$

because $\gamma - 1 < 0$ and $\lim_{\alpha \rightarrow 0} \alpha^{\gamma-1} = +\infty$. Thus the FOC is satisfied for any $\alpha_{\ell,k}^* = 0$.

Step 3: Interior solutions. For $\alpha_{\ell,k}^* > 0$, we have $\mu_{\ell,k} = 0$ and the FOC yields:

$$\alpha_{\ell,k}^* = \left(\frac{\gamma \tau_{\ell,k}^0}{\lambda_\ell} \right)^{\frac{1}{1-\gamma}}.$$

This is increasing in $\tau_{\ell,k}^0$ and decreasing in λ_ℓ . Rearranging gives Equation (12).

Step 4: Determining the active set. Let signals be ordered by precision: $\tau_{\ell,1}^0 \geq \tau_{\ell,2}^0 \geq \dots \geq \tau_{\ell,K_\ell}^0$. Define the threshold precision:

$$\bar{\tau}(\lambda_\ell) = \frac{\lambda_\ell}{V'_\ell(\tau_\ell^*) \gamma} \cdot A_\ell^{1-\gamma}.$$

Then $\alpha_{\ell,k}^* > 0$ if and only if $\tau_{\ell,k}^0 > \bar{\tau}(\lambda_\ell)$. The active set is $\mathcal{A}_\ell = \{k : \tau_{\ell,k}^0 > \bar{\tau}\}$. Because $\bar{\tau}$ decreases in A_ℓ , more signals become active as the budget grows.

Step 5: Uniqueness. The objective is strictly concave (composition of strictly increasing concave V_ℓ with strictly concave precision function), and the constraint set is convex. A strictly concave function on a convex set has a unique maximizer. $\square$

B Proof of Proposition 2 (Distortion Comparative Statics)

Recall $\Delta_\ell = 1 - \rho_\ell$, where ρ_ℓ is given by Definition 4. For the bottom layer with $\tau_\ell^{\text{prior}} = 0$, the transmission factor reduces to:

$$\rho_\ell = \frac{\sum_k g_\ell(\alpha_{\ell,k}^*) \tau_{\ell,k}^0}{\sum_k \tau_{\ell,k}^0}.$$

(i) Budget effect. From the KKT solution (Proposition 1), interior attention weights satisfy $\alpha_{\ell,k}^* = (\gamma\tau_{\ell,k}^0/\lambda_\ell)^{1/(1-\gamma)}$. Summing over the active set and imposing the budget constraint yields:

$$\lambda_\ell = \gamma \left(\frac{\sum_{k \in \mathcal{A}_\ell} (\tau_{\ell,k}^0)^{\frac{1}{1-\gamma}}}{A_\ell} \right)^{1-\gamma}.$$

Substituting back, the optimal weight is:

$$\alpha_{\ell,k}^* = A_\ell \cdot \frac{(\tau_{\ell,k}^0)^{\frac{1}{1-\gamma}}}{\sum_{j \in \mathcal{A}_\ell} (\tau_{\ell,j}^0)^{\frac{1}{1-\gamma}}}, \quad k \in \mathcal{A}_\ell.$$

Hence $\alpha_{\ell,k}^* = c_k A_\ell$ where $c_k > 0$ depends only on the relative precisions. With $g_\ell(\alpha) = \alpha^\gamma$, we have $g_\ell(\alpha_{\ell,k}^*) = (c_k A_\ell)^\gamma$. The transmission factor becomes:

$$\rho_\ell = \frac{A_\ell^\gamma \sum_{k \in \mathcal{A}_\ell} c_k^\gamma \tau_{\ell,k}^0}{\sum_{k=1}^{K_\ell} \tau_{\ell,k}^0},$$

which is increasing in A_ℓ . Therefore $\partial \Delta_\ell / \partial A_\ell = -\partial \rho_\ell / \partial A_\ell < 0$.

(ii) Signal count effect. When K_ℓ increases while A_ℓ is fixed, the budget is spread more thinly across signals. The marginal signals added have lower intrinsic precision, reducing the average precision per signal. The transmission factor ρ_ℓ decreases because the denominator $\sum_k \tau_{\ell,k}^0$ increases (holding average precision constant) while the numerator increases more slowly due to the concavity of g_ℓ .

(iii) Heterogeneity effect. Consider two precision distributions with the same mean $\bar{\tau}^0$ but different variances. Under uniform attention $\alpha_{\ell,k}^* = A_\ell/K_\ell$ (optimal for identical signals), $\rho_\ell^{\text{uniform}} = (A_\ell/K_\ell)^\gamma$. Under a heterogeneous distribution with a dominant signal $k = 1$, the optimal allocation concentrates attention: $\alpha_{\ell,1}^* > A_\ell/K_\ell > \alpha_{\ell,k}^*$ for $k > 1$. Because $g(\cdot)$ is concave, Jensen's inequality implies $\sum_k g(\alpha_{\ell,k}^*) \tau_{\ell,k}^0$ is higher under heterogeneous concentration, holding $\sum_k \tau_{\ell,k}^0$ fixed. Thus $\rho_\ell^{\text{heterogeneous}} > \rho_\ell^{\text{uniform}}$ and Δ_ℓ is lower. $\square$

C Proof of Proposition 4

From the recursive structure (Equation (28)), the top-layer precision satisfies:

$$\tau_L^* = \rho_{L-1} \cdot \tau_{L-1}^* + \sum_{k=1}^{K_L} g_L(\alpha_{L,k}^*) \tau_{L,k}^0,$$

with $\tau_0^* = \tau_0^0$ and $\rho_\ell \in [0, 1)$ for any finite A_ℓ . Define the first difference $\Delta \tau_L^* \equiv \tau_L^*(L+1) - \tau_L^*(L)$.

Adding a layer increases the chain length but also adds a new precision source. The net effect is:

$$\Delta \tau_L^* = \underbrace{\rho_L \cdot \tau_L^*}_{\text{updated inherited precision}} + \underbrace{\sum_{k=1}^{K_{L+1}} g_{L+1}(\alpha_{L+1,k}^*) \tau_{L+1,k}^0}_{\text{new layer's contribution}} - \tau_L^*.$$

Since $\rho_L < 1$, the first term is less than τ_L^* . The second term is bounded above by the first-best precision. For any interior solution (finite A_{L+1}), the loss from compression exceeds the gain from the new layer, so $\Delta\tau_L^* < 0$.

For the second difference, each additional layer reduces both the inherited precision and the marginal contribution of the new layer. Hence:

$$\Delta\tau_L^*(L+1) - \Delta\tau_L^*(L) < 0,$$

which is the discrete analog of the second derivative being negative. $\square$

D Proof of Proposition 5 (Cost-Minimizing Budget)

Step 1: Define the objective. Let $\Pi(L) = V_L(\tau_L^*) - \sum_{\ell=0}^{L-1} c_\ell(A_\ell)$ denote the principal's net value when the chain has L layers.

Step 2: Concavity of τ_L^* . From Proposition 7, the first difference $\Delta\tau_L^* \equiv \tau_L^*(L+1) - \tau_L^*(L) < 0$, and the second difference is also negative. Hence τ_L^* is strictly concave on the discrete domain.

Step 3: Concavity of $V_L(\tau_L^*)$. The value function $V_L(\cdot)$ is strictly increasing and strictly concave. The composition of a strictly increasing concave function with a strictly concave function is strictly concave. Hence $V_L(\tau_L^*(L))$ is strictly concave in L .

Step 4: Convexity of costs. The cost term $C(L) = \sum_{\ell=0}^{L-1} c_\ell(A_\ell)$ is linear in L when budgets are fixed. If budgets increase with depth, the cost is convex because $c_\ell(\cdot)$ is convex.

Step 5: Sum of concave functions. The sum of strictly concave functions is strictly concave. Hence $\Pi(L) = V_L(\tau_L^*(L)) - C(L)$ is strictly concave on $\{0, 1, 2, \dots\}$.

Step 6: Boundary behavior. At $L = 0$, $\Pi(0) = V_0(\tau_0^0) > -\infty$. As $L \rightarrow \infty$, $\tau_L^* \rightarrow 0$ because $\rho_\ell < 1$ and the product $\prod_{\ell=0}^{L-1} \rho_\ell \rightarrow 0$ geometrically. Hence $V_L(\tau_L^*) \rightarrow V_L(0)$, while $C(L) \rightarrow +\infty$. Therefore $\lim_{L \rightarrow \infty} \Pi(L) = -\infty$.

Step 7: Existence of unique maximizer. A strictly concave function on a discrete domain with $\Pi(0) > -\infty$ and $\lim_{L \rightarrow \infty} \Pi(L) = -\infty$ has a unique finite maximizer. $\square$

E Proof of Proposition 9 (Budget and Depth)

Step 1: Discrete supermodularity. Let $\Pi(L; A_\ell) \equiv V_L(\tau_L^*(A_\ell)) - \sum_{j=0}^{L-1} c_j(A_j)$ denote the principal's net value. We show that Π is supermodular in (L, A_ℓ) on $\mathbb{N} \times \mathbb{R}_+$.

Step 2: Decompose the difference. For any $L \geq 0$ and $A'_\ell > A_\ell$:

$$\begin{aligned} \Delta_L \Pi(L; A'_\ell) - \Delta_L \Pi(L; A_\ell) &= [V_{L+1}(\tau_{L+1}^*(A'_\ell)) - V_{L+1}(\tau_{L+1}^*(A_\ell))] \\ &\quad - [V_L(\tau_L^*(A'_\ell)) - V_L(\tau_L^*(A_\ell))]. \end{aligned}$$

Step 3: Sign of each term. The top-layer precision with $L+1$ layers is:

$$\tau_{L+1}^*(A_\ell) = \rho_L(A_\ell) \cdot \tau_L^* + \sum_{k=1}^{K_{L+1}} g_{L+1}(\alpha_{L+1,k}^*) \tau_{L+1,k}^0.$$

From Proposition 2, $\partial \rho_\ell / \partial A_\ell > 0$. Hence $\tau_{L+1}^*(A'_\ell) > \tau_{L+1}^*(A_\ell)$, and since $V_{L+1}(\cdot)$ is increasing, term (i) > 0 .

Term (ii) is similarly positive: a larger budget at layer ℓ raises the precision transmitted to all downstream layers.

Step 4: Compare the gains. The budget increase at layer ℓ affects all layers $j > \ell$ through the transmission chain. At layer $L + 1$, the effect is amplified by the additional layer of compounding. Because both V_{L+1} and V_L are increasing and concave, and the precision gain is larger at $L + 1$ than at L , the difference (i) $-$ (ii) > 0 .

Step 5: Apply Topkis's theorem. The cost term does not depend on A_ℓ , so it does not affect the cross-difference. Since $\Delta_L \Pi(L; A'_\ell) > \Delta_L \Pi(L; A_\ell)$, the objective is supermodular in (L, A_ℓ) . By Topkis's monotonicity theorem, $L^*(A_\ell) \equiv \arg \max_L \Pi(L; A_\ell)$ is increasing in A_ℓ . $\square$

F Proof of Proposition 11 (Cost and Depth)

Step 1: Discrete submodularity. Let $\Pi(L; \kappa_\ell) \equiv V_L(\tau_L^*) - \sum_{j=0}^{L-1} c_j(A_j; \kappa_\ell)$, where $c_j(A_j; \kappa_\ell) = \kappa_\ell A_\ell + \frac{\phi_\ell}{2} A_\ell^2$. We show that Π is submodular in (L, κ_ℓ) .

Step 2: Cost effect on the marginal value. The difference in marginal values is:

$$\begin{aligned} \Delta_L \Pi(L; \kappa'_\ell) - \Delta_L \Pi(L; \kappa_\ell) &= -[c_\ell(A_\ell; \kappa'_\ell) - c_\ell(A_\ell; \kappa_\ell)] \cdot \mathbf{1}_{\{\ell < L+1\}} \\ &= -(\kappa'_\ell - \kappa_\ell) A_\ell \cdot \mathbf{1}_{\{\ell < L+1\}} \leq 0. \end{aligned}$$

For $\ell < L + 1$, a higher cost reduces the marginal net value of adding the $(L + 1)$ -th layer.

Step 3: Monotonicity. Because $\Delta_L \Pi(L; \kappa'_\ell) \leq \Delta_L \Pi(L; \kappa_\ell)$, the objective is submodular in (L, κ_ℓ) . By Topkis's theorem, $L^*(\kappa_\ell) \equiv \arg \max_L \Pi(L; \kappa_\ell)$ is decreasing in κ_ℓ . $\square$

G Proof of Proposition 12 (Irreducibility of Incentive Friction)

This appendix provides the complete proof of Proposition 12, which establishes the finiteness of optimal depth under diminishing signal quality. The proof proceeds in five logical steps.

Step 1: Recursive structure. Under uniform technology ($g_\ell = g$, $c_\ell = c$, $K_\ell = K$ for all ℓ) and Assumption 8, the equilibrium posterior precision at layer ℓ evolves according to:

$$\tau_\ell^* = \underbrace{\rho \cdot \tau_{\ell-1}^*}_{\text{precision carried forward}} + \underbrace{\sum_{k=1}^K g(\alpha_k^*) \tau_{\ell,k}^0}_{\text{new signal contribution}}. \quad (\text{A.7.1})$$

By Assumption 8, $\tau_{\ell,k}^0 = \tau_k^0 \eta^\ell$, so the new signal contribution becomes:

$$\sum_{k=1}^K g(\alpha_k^*) \tau_k^0 \eta^\ell = C \eta^\ell, \quad (\text{A.7.2})$$

where C is defined in (36). The initial condition is $\tau_0^* = \sum_{k=1}^K \tau_{0,k}^0 = \sum_{k=1}^K \tau_k^0$. Thus (A.7.1) simplifies to:

$$\tau_\ell^* = \rho \tau_{\ell-1}^* + C \eta^\ell, \quad \tau_0^* \text{ given}. \quad (\text{A.7.3})$$

Step 2: Closed-form solution. Equation (A.7.3) is a first-order linear non-homogeneous difference equation with varying coefficients in the forcing term.

Case 1: $\eta \neq \rho$. Decompose $\tau_\ell^* = \tau_\ell^{(h)} + \tau_\ell^{(p)}$. The homogeneous solution is $\tau_\ell^{(h)} = A\rho^\ell$, and the ansatz $\tau_\ell^{(p)} = B\eta^\ell$ yields:

$$B\eta^\ell = \rho B\eta^{\ell-1} + C\eta^\ell \implies B\eta = \rho B + C\eta \implies B = \frac{C\eta}{\eta - \rho}.$$

Applying the initial condition $\tau_0^* = A + B$:

$$A = \tau_0^* - \frac{C\eta}{\eta - \rho}. \quad (\text{A.7.4})$$

Therefore, for all $L \geq 0$:

$$\tau_L^* = \rho^L \tau_0^* + \frac{C\eta}{\eta - \rho} (\eta^L - \rho^L), \quad (\text{A.7.5})$$

which coincides with equation (37) in the main text.

Case 2: $\eta = \rho$. The recurrence becomes $\tau_\ell^* = \eta \tau_{\ell-1}^* + C\eta^\ell$. Dividing both sides by $\eta^{\ell\ell}$:

$$\frac{\tau_\ell^*}{\eta^\ell} = \frac{\tau_{\ell-1}^*}{\eta^{\ell-1}} + C. \quad (\text{A.7.6})$$

Defining $u_\ell \equiv \tau_\ell^*/\eta^\ell$, this becomes $u_\ell = u_{\ell-1} + C$, which telescopes to $u_\ell = u_0 + C\ell = \tau_0^* + C\ell$. Restoring τ_L^* :

$$\tau_L^* = \eta^L (\tau_0^* + CL). \quad (\text{A.7.7})$$

Step 3: Vanishing precision (Part (i)). We establish that $\lim_{L \rightarrow \infty} \tau_L^* = 0$ in all cases.

For $\eta \neq \rho$, since $\rho, \eta \in (0, 1)$:

$$\lim_{L \rightarrow \infty} \rho^L = 0, \quad \lim_{L \rightarrow \infty} \eta^L = 0, \quad (\text{A.7.8})$$

so both terms in (A.7.5) converge to zero. For $\eta = \rho$, the linear growth of $\tau_0^* + CL$ is asymptotically dominated by the exponential decay of η^L :

$$\lim_{L \rightarrow \infty} \eta^L (\tau_0^* + CL) = 0, \quad (\text{A.7.9})$$

since $\lim_{L \rightarrow \infty} L^m \eta^L = 0$ for any $m \geq 0$ and $\eta \in (0, 1)$. This completes Part (i).

Step 4: Eventual monotonic decline (Part (ii)). From (A.7.3), the increment is:

$$\Delta_\ell \equiv \tau_\ell^* - \tau_{\ell-1}^* = -(1 - \rho)\tau_{\ell-1}^* + C\eta^\ell. \quad (\text{A.7.10})$$

We prove $\Delta_\ell < 0$ for all sufficiently large ℓ by considering three subcases.

Subcase 4a: $\eta > \rho$. From (A.7.5), as $\ell \rightarrow \infty$, the term η^ℓ dominates ρ^ℓ , yielding:

$$\tau_{\ell-1}^* \sim \frac{C\eta^\ell}{\eta - \rho} \quad (\ell \rightarrow \infty). \quad (\text{A.7.11})$$

Substituting into (A.7.10):

$$\Delta_\ell \sim -(1 - \rho) \frac{C\eta^\ell}{\eta - \rho} + C\eta^\ell = C\eta^\ell \left(\frac{\eta - \rho - 1 + \rho}{\eta - \rho} \right) = C\eta^\ell \frac{\eta - 1}{\eta - \rho}. \quad (\text{A.7.12})$$

Since $\eta < 1$, the numerator $\eta - 1 < 0$, while the denominator $\eta - \rho > 0$. Hence $\Delta_\ell \sim [\text{negative constant}] \times \eta^\ell < 0$ for all large ℓ .

Subcase 4b: $\eta < \rho$. Now ρ^ℓ dominates, giving:

$$\tau_{\ell-1}^* \sim \left(\tau_0^* + \frac{C\eta}{\rho - \eta} \right) \rho^{\ell-1} \quad (\ell \rightarrow \infty). \quad (\text{A.7.13})$$

Substituting into (A.7.10):

$$\Delta_\ell \sim -(1 - \rho) \left(\tau_0^* + \frac{C\eta}{\rho - \eta} \right) \rho^{\ell-1} + C\eta^\ell. \quad (\text{A.7.14})$$

Since $\rho > \eta$, the negative term decays as $\rho^{\ell-1}$ while the positive term decays as η^ℓ . The former decays strictly slower and has a negative coefficient, so it dominates for large ℓ . Thus $\Delta_\ell < 0$ for all $\ell \geq \bar{L}$ with:

$$\bar{L} = \inf \left\{ \ell \in \mathbb{N} : \ell > \frac{\ln \left(\frac{(1-\rho)}{C} \left(\tau_0^* + \frac{C\eta}{\rho - \eta} \right) \right)}{\ln(\eta/\rho)} \right\} + 1, \quad (\text{A.7.15})$$

which is finite since $\ln(\eta/\rho) < 0$.

Subcase 4c: $\eta = \rho$. Using (A.7.7):

$$\begin{aligned} \Delta_\ell &= \eta^\ell (\tau_0^* + C\ell) - \eta^{\ell-1} (\tau_0^* + C(\ell-1)) \\ &= \eta^{\ell-1} \left[\underbrace{(\eta-1)\tau_0^*}_{<0} + \underbrace{C(1-(1-\eta)\ell)}_{\rightarrow -\infty} \right]. \end{aligned} \quad (\text{A.7.16})$$

Since the bracket is affine in ℓ with negative slope $-C(1-\eta) < 0$, there exists:

$$\bar{L} = \left\lfloor \frac{\tau_0^*}{C} + \frac{1}{1-\eta} \right\rfloor + 2, \quad (\text{A.7.17})$$

such that $\Delta_\ell < 0$ for all $\ell \geq \bar{L}$. This completes Part (ii).

Step 5: Finite optimal depth (Part (iii)). Strict monotonicity of $V(\cdot)$ and Part (ii) together imply:

$$V(\tau_\ell^*) < V(\tau_{\ell-1}^*) \quad \text{for all } \ell \geq \bar{L}. \quad (\text{A.7.18})$$

Thus $\{V(\tau_\ell^*)\}_{\ell=\bar{L}}^\infty$ is strictly decreasing beyond $\bar{L}$, so its maximum is attained at some $L^* \in \{0, 1, \dots, \bar{L}\}$.

When the marginal attention cost is $\kappa \geq 0$, adding any layer $\ell \geq \bar{L}$ yields net marginal benefit:

$$\underbrace{V(\tau_\ell^*) - V(\tau_{\ell-1}^*)}_{<0} - \kappa < 0, \quad (\text{A.7.19})$$

which is strictly negative. Hence no planner would ever choose $\ell > \bar{L}$, and $L^* \leq \bar{L} < \infty$ regardless of how small κ is. This completes the proof of Proposition 12. $\blacksquare$

Remark 5 (Comparative statics on $\bar{L}$). *The threshold $\bar{L}$ beyond which precision decreases is increasing in the baseline precision τ_0^* and decreasing in the signal contribution C and the attenuation factor η . Specifically, a higher η (better signal preservation) pushes $\bar{L}$ outward, allowing more layers before returns turn negative. In the limit $\eta \rightarrow 1$, the threshold diverges, recovering the original model where infinite depth can be optimal.*

Remark 6 (Comparison to the no-attenuation case). *When $\eta = 1$ (no signal degradation), the recurrence (A.7.3) becomes $\tau_\ell^* = \rho\tau_{\ell-1}^* + C$, with solution $\tau_L^* = \rho^L\tau_0^* + C(1 - \rho^L)/(1 - \rho) \rightarrow C/(1 - \rho) > 0$. The precision converges to a positive limit, so $V(\tau_L^*)$ converges to $V(C/(1 - \rho)) > V(0)$ from below, and infinite depth is optimal whenever $\kappa < V(C/(1 - \rho)) - \lim_{L \rightarrow \infty} V(\tau_{L-1}^*) = 0^+$. The diminishing signal quality assumption ($\eta < 1$) is therefore necessary and sufficient for the finite-depth result under uniform technology.*

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